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Resource Allocation and Robust Trajectory Design for Dual UAV-Assisted Secure Communications

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***Abstract*—This paper introduces a dual unmanned aerial vehicle (UAV)-assisted communication system to facilitate secure information transfer between an Internet of Things (IoT) device and a base station in the presence of multiple eavesdroppers. We consider a more realistic scenario where the legitimate nodes have imperfect knowledge about the eavesdroppers' locations that can intercept information through air-to-ground and ground-to-air links. Both UAVs and energy-constrained IoT device harvest energy from radio-frequency signals transmitted by a nearby power beacon. Utilizing the harvested energy, one UAV acts as an aerial relay while the other UAV broadcasts jamming signals to ensure information secrecy. In the considered system, we formulate an average secrecy rate maximization problem by jointly optimizing the time-splitting ratio, transmit powers of IoT device and UAVs, and trajectories of both UAVs. Moreover, we account for practical constraints such as collision avoidance, limited battery capacity, energy harvesting and information causality, and UAVs' maximum velocity and flying radius. The formulated problem is non-smooth and non-convex, which we first transform into a smooth objective function without affecting its optimality. Then, we relax the eavesdropper's uncertainty constraint by using Jensen's inequality and obtain upper bound of the wiretapping rate. Further, we decompose the original optimization problem into five subproblems using a block coordinate descent algorithm and transform them into convex form via successive convex approximation. These subproblems are then solved using the convex optimization tool CVX, simplex, and subgradient descent methods. Lastly, we show the effectiveness of our framework by comparing the secrecy performance with benchmark schemes in simulation results.**

***Index Terms*—Unmanned aerial vehicle, UAV relaying, cooperative jamming, average secrecy rate, trajectory optimization.**

I. Introduction

The rapid advancement of wireless technology promises seamless connectivity and communication, embracing a new era of wireless experiences [1]. Due to this, the demand for ultra-reliable and ubiquitous communications has led to the immense development of unmanned aerial vehicle (UAV)-assisted communications [2]. With the evolving beyond fifth-generation standards, UAVs are expected to become the future hot spot for providing ubiquitous connectivity owing to their line-of-sight (LoS)-dominant air-to-ground (A2G) and ground-to-air (G2A) links and flexible mobility. Moreover, integrating UAVs and Internet of Things (IoT) devices offers promising solutions for various vital applications, including remote monitoring, intelligent transportation, smart agriculture, sites with poor or no connectivity, and natural or artificial disasters. UAVs operating as aerial nodes can collect data from either remote-located IoT devices or from those in overcrowded and highly shadowed areas [3].

On the other hand, a critical concern for UAV-aided communications arises due to their broadcast nature, which makes them susceptible to wiretapping by malicious nodes. The intruders can exploit UAV's A2G channels to access confidential information [4]. Although high-layer cryptographic protocols are used for data security, the encryption operations usually introduce processing delays, which are unsuitable for real-time applications [5]. Moreover, they demand high computational complexity, resulting in significant energy consumption, which is unsustainable for UAV systems. Over the last few years, physical layer security (PHY security) has been introduced to safeguard wireless networks against intruders by leveraging communication channel conditions [6]. Various PHY security techniques and cooperative methods, including cooperative jamming and artificial noise (AN) injection have proven effective in securing UAV-assisted communications [7].

Recent research has focused on enhancing PHY security in UAV-assisted networks through optimal trajectory design [8]–[12]. By flying closer to legitimate nodes and away from eavesdroppers, UAVs reduce the risk of interception. Specifically, the authors in [8]–[11] employed block coordinate descent (BCD) and successive convex approximation (SCA) algorithms to maximize secrecy-related metrics, such as average secrecy rate (ASR), secure computation, and secrecy capacity, in their respective models. Whereas the authors in [12] employed alternating optimization (AO) and SCA-based iterative methods to maximize secrecy throughput. Incorporating friendly jammer UAVs can offer additional security by broadcasting interference signals to degrade the wiretap link [13]–[21]. The authors in [13]–[16] focused on maximizing the secrecy performance by optimizing the power allocations and trajectories of both relay and jammer UAVs. Whereas the authors in [17] considered some practical constraints, such as velocity limits and collision avoidance, in their ASR maximization problem.

The authors in [18] and [19] formulated problems for maximizing secrecy rates, considering trajectory-related and information-causal constraints in their respective dual UAV-

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assisted secured networks. However, energy causality constraints have not been considered in [13]–[19]. Moreover, these studies, [8]–[19], have assumed perfect knowledge of the eavesdropper's locations, which is impractical in real-world applications. In contrast, the authors in [20] maximized secrecy energy efficiency by considering multiple eavesdroppers at imperfect locations. Whereas the authors in [21] modeled the secrecy rate maximization problem for mobile ground nodes and eavesdroppers using a Markov decision process.

Another challenge arises due to the energy constraints of the battery-powered IoT devices and UAVs that restrict their operational longevity [22]. Regardless of the implementation of energy-saving protocols, the IoT devices operate over a limited period, thus mandating frequent recharging or replacement of their batteries [23]. UAVs, operating as aerial relays or base stations, require additional energy for relaying operations. However, their limited size, weight, and power constrain the capacity of onboard batteries, impacting their flight duration and overall operational time [24]. Over the past few years, the wireless power transfer (WPT) technique has shown great potential for overcoming the battery capacity limitations of UAVs and IoT devices [25]. Among various energy harvesting (EH) methods, radio-frequency (RF)-EH techniques have become more prevalent in enhancing wireless networks' lifetimes as they are less susceptible to adverse effects from weather and environmental factors [26]. To address the limited battery challenge, RF-EH offers a promising solution to supplement onboard batteries and provide energy for data or jamming signal transmission operations [27].

Several recent studies have explored WPT and PHY security in UAV communications to enhance network longevity and security [28]–[33]. For instance, the authors in [28] solved non-convex secrecy rate (SR) maximization by jointly optimizing UAV trajectory, transmit power, and wireless charging duration. Whereas the authors in [29] maximized uplink throughput in UAV-assisted data collection in the presence of an eavesdropper with imperfect location. Similarly, the authors in [30] and [31] considered imperfect locations of eavesdroppers and Poisson-distributed eavesdroppers, respectively, and maximized the SR. The authors in [32] extended this approach to UAV-assisted cognitive radio systems. However, these works addressed the security concerns either in A2G or G2A links, taking a single UAV in their model. In contrast, the authors in [33] utilized a dual-UAV system to maximize worst-case SR. Based on the literature discussion outlined above, the following research gaps can be identified:

- Various research studies [8]–[21], [28], [29], [31], [32] have addressed concerns regarding information secrecy in either in A2G or G2A links.
- The authors in [8]–[21] did not incorporate the RF-EH technique into their models.
- The works in [8]–[19], [28] assumed perfect knowledge of eavesdroppers' locations, which does not reflect a realistic scenario.
- The authors in [13]–[19], [28], [29], [31], [32] did not thoroughly consider various practical constraints, such as collision avoidance, the operational radius of UAVs, and energy and information causality.
- Other works [18], [28], [29], [32] did not take the EH parameter into account in their optimization problems.

Motivated by the existing literature gaps, we utilize a dual UAV-assisted relaying and cooperative jamming scheme for secure IoT communications network. Different from the studies in [8]–[21], we enable EH at the IoT device and UAVs using the RF signals received from a nearby power beacon. The IoT device utilizes the harvested energy to transmit the data to a dedicated receiver via a mobile UAV relay. Additionally, we explore a worst scenario involving multiple eavesdroppers at uncertain locations that can wiretap both A2G and G2A communication links. Unlike studies such as [8]–[12], [28], we utilize a dual UAV system comprising a jammer UAV to transmit jamming signals, thereby confusing eavesdroppers. A brief overview of the paper's significant contributions are:

- For this setup, we formulate an ASR maximization problem by jointly optimizing the TSR, transmit powers of IoT device, relay and jammer UAVs, and trajectories of these UAVs. Different from [13]–[19], [28], [29], [31], [32], we consider practical constraints, such as collision avoidance, UAVs' maximum radius, peak and average powers, UAVs' limited battery, and information and energy causality.
- The formulated problem is non-smooth and non-convex with tightly coupled variables. So, we first tackle the non-smoothness of the original problem without compromising its optimality and decompose it into five subproblems. Then, we employ the SCA method to address the non-convexity and convert it into tractable forms.
- Following this, we devise an iterative algorithm utilizing the BCD method to solve the subproblems alternately and compute numerical solutions for the optimizing variables. Then, we verify the efficacy of our proposed scheme by comparing it with benchmark schemes and highlight the valuable insights for efficient system design.

To the best of the authors' knowledge, no existing literature addresses the robust design of secure communications using dual UAVs while accounting for practical constraints. We provide Table I to highlight the contributions made in this work and compare them with relevant studies. The remaining part of the paper is organized into the following sections: The proposed model is presented in Section II-A, while we discuss the transmission scheme and obtain the expression of SR in Section II-B. We formulate the ASR maximization problem in Section III and then propose SCA and BCD algorithms to solve the subproblems efficiently in Section IV. Finally, we discuss the simulation results in Section V and conclude the paper in Section VI. Table II summarizes the list of notations that are used in this paper.

II. System Model Discussion

A. System Description

We consider a dual-UAV-assisted secure communication network, as shown in Fig. 1(a), in which an energy-constrained IoT device (S) communicates with the base station (D), however the direct path between them is heavily obstructed. So, we utilize UAV relay (U_R) to facilitate the communication

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TABLE I: Comparison of proposed work with relevant literature works.

Works	System Model					Causality Constraints	Collision Avoidance	EH Parameter Optimization	Power Allocation
	Aerial Links	Number of Eavesdroppers	UAV Jammer	Power Beacon	Eavesdroppers' Uncertain Location				
[10]	G2A	Single	×	×	×	None	×	×	✓
[11]	G2A	Single	×	×	×	Information	×	×	✓
[18]	A2G	Single	✓	×	×	Information	×	×	⊙
[28]	A2G	Single	×	✓	×	Energy	×	✓	⊙
[29]	A2G	Multiple	×	×	×	Energy	×	×	⊙
[30]	A2G and G2A	Multiple	×	×	✓	Energy	×	✓	✓
[32]	A2G	Single	✓	×	×	Energy	✓	×	⊙
[33]	A2G	Multiple	✓	×	✓	Energy	×	✓	⊙
This work	A2G and G2A	Multiple	✓	✓	✓	Both	✓	✓	✓
✓ Covered		⊙ Partially covered				× Not covered			

link between S and D. There are multiple terrestrial eavesdroppers, specifically n-th eavesdropper is denoted as E_n, $\forall n \in \{1, 2, \ldots, N\}$, present near S and D, that can wiretap the information. Therefore, a UAV jammer (U_J) is leveraged to secure the data transfer by broadcasting jamming signals to degrade the eavesdropper's wiretap link [13]–[21]. In the considered system, a power beacon (B) is used for transmitting the RF signal to the energy-constrained nodes, i.e., S, U_R, and U_J, which perform EH from the signal and utilize it for their operation [28]–[31], [33].

We employ trajectory discretization as depicted in Fig. 1(b) by dividing the UAVs' finite flight duration, denoted as T, into M time slots with equal intervals of $\delta_m = \frac{T}{M}$. It is important to choose sufficiently smaller values of δ_m, to ensure the UAVs' location remains constant from the perspective of communicating nodes within each time slot $[m]$, where $m \in \mathcal{M} \triangleq \{1, 2, \ldots, M\}$. As shown in Fig. 1(b), the data transmission block is segmented into three phases, i.e., one WPT phase and two information transfer (IT) phases. Specifically, the WPT phase is of $\alpha[m]\delta_m$ duration, where $\alpha[m] \in [0, 1]$ represents the TSR [28], [33]. Then, S transmits information to U_R in the first IT phase of $\frac{(1-\alpha[m])\delta_m}{2}$ duration. Different from the authors in [30], that utilized AF relaying, U_R decodes the S's information and forwards it to D in the second IT phase using decode and forward (DF) relaying. Moreover, U_J broadcasts jamming signals to degrade the eavesdropper's wiretap link and ensure data security in both IT phases.

Two-dimensional (2D) Cartesian coordinate system is used to describe the locations of legitimate terrestrial nodes, i.e., B, S, and D are located at $\mathbf{w}_t = (x_t, y_t)$ such that $t \in \{B, S, D\}$. We assume that the locations of the legitimate terrestrial nodes are known in advance to both UAVs [8], [15]–[17], [19]–[21], [29], [30], [34]. Moreover, $\mathbf{w}_{E_n} = (x_{E_n}, y_{E_n})$ represents the exact location of E_n, which is imperfectly known to the legitimate nodes [20], [21], [29]–[33]. It is important to note that the legitimate nodes can estimate the location of E_n, denoted as $\tilde{\mathbf{w}}_{E_n} = (\tilde{x}_{E_n}, \tilde{y}_{E_n})$, using advanced feasible techniques [17]. So, the relation between $\mathbf{w}_{E_n}$ and $\tilde{\mathbf{w}}_{E_n}$ is given as $x_{E_n} = \tilde{x}_{E_n} + \Delta x_{E_n}$ and $y_{E_n} = \tilde{y}_{E_n} + \Delta y_n$, wherein $(\Delta x_n, \Delta y_n) \in \mathbb{E}_n, \{(\Delta x_n, \Delta y_n) \mid \Delta x_n^2 + \Delta y_n^2 \leq \Delta Q_n^2\}$ and $\|\tilde{\mathbf{w}}_{E_n} - \mathbf{w}_{E_n}\| \leq \Delta Q_n$, where Δx_n and Δy_n are the estimation errors and assumed to be bounded within a circle,

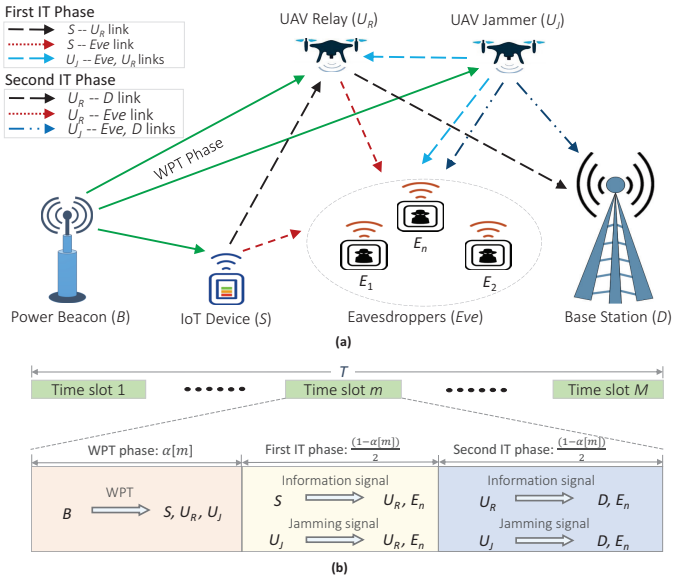

Fig. 1: (a) Dual UAV-assisted secure communications. (b) Description of the time slot framework.

$\mathbb{E}_n$ represents the possible set of errors, and the maximum estimation error's magnitude square is represented as ΔQ_n^2 [21], [29], [32], [33], [35]. Furthermore, we consider a practical assumption for the uncertain region of E_n, i.e., ΔQ_n is generally lesser than the distance between E_n and other communicating nodes [30], [33].

The coverage area of our considered dual-UAV-assisted secure communication network resembles the microcell communication scenarios with a coverage radius of approximately $100\,\mathrm{m}$ to $1000\,\mathrm{m}$, consistent with the standards specified in the 802.11, LTE, or 5G frameworks [8], [19], [34], [36], [37]. Numerous literature works have also taken similar system assumptions for their UAV-enabled communication models [15]–[17], [20], [29], [30]. For the considered dual-UAV system, the three-dimensional positions of UAVs can be expressed as $(\mathbf{q}_i[m], H_i)$, where $\mathbf{q}_i[m] = (x_i[m], y_i[m])$ and H_i are respectively the time-varying 2D positions and fixed flying altitude of both UAVs, such that $i \in \{U_R, U_J\}$. UAVs are assumed to follow trajectory with predetermined initial and final location, denoted as $\mathbf{q}_{i_1} = \mathbf{q}_i[1]$ and $\mathbf{q}_{i_F} = \mathbf{q}_i[M]$, respectively. We assume the altitudes of U_R and U_J are

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TABLE II: List of notations.

Symbol	Description
T, δ_m	Flight period of UAVs and length of each time slot
$\mathbf{q}_x[m]$	2D location of aerial node x
$\mathbf{q}_{i_1}$, $\mathbf{q}_{i_F}$	Initial and final locations of UAVs
$\mathbf{w}_x$	2D location of terrestrial node x
$\tilde{\mathbf{w}}_{E_n}$	Estimated 2D location of E_n
$r_{i,\max}$	Maximum flying radius of UAVs
$v_{i,\max}$	Maximum velocity of UAVs
$P_x[n]$, P_x^{avg}	Transmit power and average transmit power of x
$P_{i,f}[m]$	Power consumption for UAVs' flying/hovering
$\beta_i^{\max}$	Average power budgets of UAVs
$g_{xy}[m]$	Time-varying power gain between nodes x and y
ρ_0	Reference channel gain
σ_x^2	Variance at node x
$\alpha[m]$	TSR in m-th time slot
$R_{\text{wir}}^{\text{ub}}[m]$	Upper bound of wiretapping rate at E_n
ΔQ_n	Radius of maximum estimation error
ξ	Exponentially distributed random variable

Index x and y represent the involved nodes, i.e., S, D, U_i, and E_n.

sufficiently high to ensure that LoS components dominate the A2G and G2A channels, aligning with the channel models specified in 3GPP TR 36.777 [9], [10], [17], [18], [20], [29], [36]–[38]. To ensure a fair and credible comparison between the simulation results of our proposed scheme and benchmark schemes, we employ a commonly used reference distance-based channel power gain model with a reference distance of $d_0 = 1\,\text{m}$ for modeling various A2G and G2A links.

Therefore, time-varying power gains of different G2A channels between the nodes B-U_R, B-U_J, and S-U_R are expressed as $g_{Bi}[m] = \frac{\rho_0}{d_{Bi}^2[m]} = \frac{\rho_0}{\|\mathbf{w}_B - \mathbf{q}_i[m]\|^2 + H_i^2}$ and $g_{SR}[m] = \frac{\rho_0}{d_{SR}^2[m]} = \frac{\rho_0}{\|\mathbf{w}_S - \mathbf{q}_R[m]\|^2 + H_i^2}$, where ρ_0 represents the channel power gain. Similarly, the power gains of A2G channels between the nodes U_R-D and U_J-D links are expressed as $g_{iD}[m] = \frac{\rho_0}{d_{iD}^2[m]} = \frac{\rho_0}{\|\mathbf{q}_i[m] - \mathbf{w}_D\|^2 + H_i^2}$, whereas for U_R-E_n and U_J-E_n, these are obtained as $g_{iE_n}[m] = \frac{\rho_0}{d_{iE_n}^2[m]} = \frac{\rho_0}{\|\mathbf{q}_i[m] - \mathbf{w}_{E_n}\|^2 + H_i^2}$. Next, we consider small-scale Rayleigh fading for the terrestrial channels, i.e., B-S and S-E_n. Therefore, the channel power gain corresponding to these terrestrial links, i.e., $g_{BS} = \frac{\rho_0}{d_{BS}^2} = \frac{\rho_0}{\|\mathbf{w}_B - \mathbf{w}_S\|^2}\xi$ and $g_{SE_n} = \frac{\rho_0}{d_{SE_n}^2} = \frac{\rho_0}{\|\mathbf{w}_S - \mathbf{w}_{E_n}\|^2}\xi$ where ξ is an exponentially distributed random variable. Additionally, energy consumption is an indispensable factor affecting the UAV's endurance; therefore, we incorporate it into the considered model [28]. Here, we consider UAVs' power consumption for flying, which is given as $P_{i,f}[m] = \frac{W_i (v_i[m])^2}{2}$ and powered by the UAVs' onboard battery, where W_i is UAVs' weight [14], [15], [21].

B. Wireless Charging and Data Transmission

S, U_R, and U_J harvest energy from the signal transmitted by B and store the collected energy in their rechargeable batteries for subsequent use in the IT phases. It is important to note that U_R and U_J utilize the harvested energy for relaying operation or broadcasting jamming signals [33]. Energy harvested by S, U_R and U_J in WPT phase is given as $\mathcal{E}_{\check{i}} = \sum_{m=1}^{M} \eta P_B[m] g_{B\check{i}} \alpha[m] \delta_m \;\forall \check{i} \in \{S, U_R, U_J\}$, where P_B represents B's transmit power, and $\alpha[m]\delta_m$ denotes the wireless charging duration, and η is the energy conversion efficiency of EH circuitry. Therefore, the maximum powers available at these nodes by performing EH are expressed as $P_{S,\max}[m] = \frac{2\eta P_B[m] g_{BS} \alpha[m]}{1-\alpha[m]}$, $P_{R,\max}[m] = \frac{2\eta P_B[m] g_{BR} \alpha[m]}{1-\alpha[m]}$, and $P_{J,\max}[m] = \frac{\eta P_B[m] g_{BJ} \alpha[m]}{1-\alpha[m]}$. In the first IT phase, S transmits the information signal to U_R, which can be overheard by E_n. Thus, we employ U_J to transmit jamming signal to degrade the E_n's wiretap link. Based on this, the signals received by U_R and E_n are given as

$$u_{SR}[m] = \sqrt{P_S[m]\; g_{SR}[m]}\; z_S[m] + \sqrt{P_J[m]\; g_{JR}[m]}\; z_J[m] + n_R[m], \tag{1}$$

$$u_{SE_n}[m] = \sqrt{P_S[m]\; g_{SE_n}}\; z_S[m] + \sqrt{P_J[m]\; g_{JE_n}[m]}\; z_E[m] + n_{E_n}[m], \tag{2}$$

where P_S, P_J and z_S, z_J denote the transmit powers and unit energy symbols of S and U_J, respectively. The additive white Gaussian noise (AWGN) at U_R and E_n are given as $n_R \sim \mathcal{CN}(0, \sigma_R^2)$ and $n_{E_n} \sim \mathcal{CN}(0, \sigma_{E_n}^2)$. It is important to note that U_R can remove the jamming signal, whereas E_n cannot identify the jamming signal while extracting the information [14], [15], [21]. The signal-to-noise ratios (SNRs) for S-U_R and S-E_n links are given as

$$\gamma_{SR}[m] = \frac{P_S[m]\; g_{SR}[m]}{\sigma_R^2}, \tag{3}$$

$$\gamma_{SE_n} = \frac{P_S[m]\; g_{SE_n}}{P_J[m] g_{JE_n}[m] + \sigma_{E_n}^2}. \tag{4}$$

Subsequently, U_R decodes the information received from S and forwards it to D in the second IT phase, which nearby eavesdroppers can wiretap. Therefore, U_J transmits the jamming signals to E_n for degrading the wiretap link. The signal received by D and E_n from U_R are given as

$$u_{RD}[m] = \sqrt{P_R[m]\; g_{RD}[m]}\; z_R[m] + \sqrt{P_J[m]\; g_{JD}[m]}\; z_J[m] + n_D[m], \tag{5}$$

$$u_{RE_n}[m] = \sqrt{P_R[m]\; g_{RE_n}[m]}\; z_R[m] + \sqrt{P_J[m]\; g_{JE_n}[m]}\; z_J[m] + n_{E_n}[m], \tag{6}$$

where z_R and P_R denote unit energy symbol and transmit power of U_R, respectively, and $n_D \sim \mathcal{CN}(0, \sigma_D^2)$ is the AWGN at D. It can be noted that D can remove the jamming signal from (5), whereas E_n is unable to identify the jamming signal in (6) while extracting the information. Therefore, the SNRs for U_R-D and U_R-E_n links are given as

$$\gamma_{RD}[m] = \frac{P_R[m]\; g_{RD}[m]}{\sigma_D^2}, \tag{7}$$

$$\gamma_{RE_n}[m] = \frac{P_R[m]\; g_{RE_n}[m]}{P_J[m] g_{JE_n}[m] + \sigma_{E_n}^2}. \tag{8}$$

We consider worst eavesdropping situation, where colluding eavesdroppers utilize a maximal-ratio combining scheme to intercept information from S and U_R in both IT phases [31]. Therefore, we express the SR as

$$R_{\text{sec}}[m] = \left[R_{\text{leg}}[m] - \max_{\Delta x_n, \Delta y_n \in \mathbb{E}_n} R_{\text{wir}}[m] \right]^+, \tag{9}$$

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where, $[x]^+ \triangleq \max\{x, 0\}$, the achievable rate from legitimate S-D links in time slot m is given by $R_{\text{leg}}[m] = \frac{(1-\alpha[m])\delta_m}{2}\log_2\left(1+\min(\gamma_{SR}[m], \gamma_{RD}[m])\right)$, and the eavesdropping rate in both IT phases is given by $R_{\text{wir}}[m] = \frac{(1-\alpha[m])\delta_m}{2}\log_2\left(1+\sum_{n=1}^{N}(\gamma_{SE_n}[m]+\gamma_{RE_n}[m])\right)$.

III. ASR Maximization Problem

In this section, we formulate ASR maximization problem for IT from S-D by jointly optimizing TSR, trajectories of U_R and U_J with transmit powers of S, U_R, and U_J (α-TRJ-PIRJ). We consider various practical constraints, such as energy causality, average power, peak power, information causality, UAV's trajectories, mobility and operation (including flight radius and collision avoidance), and TSR constraints, while formulating the problem (P1), given in (10a). Specifically, we provide *energy causality constraint* in (10b), (10c), and (10d). According to this, total energy expenditure for data transfer through S and U_R or jamming signal transmission from U_J should be less than or equal to the total harvested energy to avoid an energy deficit [28], [30], [31], [33].

Focusing on a generic scenario, we take both *average and peak power constraints* subject to P_S, P_R and P_J, respectively [9], [11]. In this regard, (10e) limits the average transmit power of S, where $P_S^{\text{avg}}[m] \leq P_{S,\max}$ and $P_S^{\text{avg}}[m] = \frac{1}{M}\sum_{m=1}^{M} P_S[m]$ is the average transmit power of S. Similar to [16], [20], we consider (10f) that constrains the battery capacities of both UAVs. Wherein $\beta_i^{\max}$ denotes the average power budget of the UAVs. In these constraints, $P_i^{\text{avg}}[m] \leq P_{i,\max}$, where $P_i^{\text{avg}}[m] = \frac{1}{M}\sum_{m=1}^{M} P_i[m]$ is the average transmit powers of UAVs. Whereas (10g) and (10h) define the peak power constraints subject to P_S, P_R, and P_J. Moreover, the *information causality constraint*, denoted by (10i), must satisfy the relation $\sum_{m=1}^{M} R_{SR}[m] \geq \sum_{m=1}^{M} R_{RD}[m],\ \forall n \in \{2, \cdots M\}$, where $R_{SR}[m] = \frac{(1-\alpha[m])\delta_m}{2}\log_2\left(1+\gamma_{SR}[m]\right)$ and $R_{RD}[m] = \frac{(1-\alpha[m])\delta_m}{2}\log_2\left(1+\gamma_{RD}[m]\right)$ are achievable rates (in bps/Hz) from $S-U_R$ and U_R-D.

Considering the *operation constraint* on U_R and U_J, (10j) ensures they reach their final locations within the allotted time. Additionally, we consider the *maximum flight radius* of UAVs permissible for their operations, denoted as $r_{i,\max}$, where (10k) lays the constraint on the flight radius of UAVs, which is less than or equal to $r_{i,\max}$. Furthermore, *collision avoidance* is crucial when designing the trajectories of multiple UAVs. We define the minimum distance between U_R and U_J, i.e., $d_{\min}$, which must be maintained to avoid collision [11], [21], [34]. The collision avoidance is considered in constraint (10l). For achieving better channel conditions, U_R and U_J can change their positions with the maximum allowable displacement and flight speed of $\hat{D}_{i,\max}$ and $v_{i,\max}$, respectively [8], [10], [11], [16], [17], [20]. Then, (10m) guarantees that U_i reaches their final positions within T. According to this *mobility constraint*, U_R and U_J can adjust their velocity with maximum velocity and acceleration, denoted as $v_{i,\max}[m]$ and $a_{i,\max}[m]$, respectively. Lastly, (10n) is the constraint on wireless charging duration for S, U_R, and U_J in the WPT phase. Based on these discussions, the ASR maximization problem is formulated as

$$\text{(P1)}: \max_{\substack{\alpha[m], P_i[m], \mathbf{q}_i[m] \\ \forall \tilde{i} \in \{S,R,J\}, \forall i \in \{R,J\}, \forall m}} \frac{1}{M}\sum_{m=1}^{M} R_{\text{sec}}[m] \tag{10a}$$

$$\textbf{s.t.} \quad \mathcal{E}_S \geq \sum_{m=1}^{M}\left(\frac{(1-\alpha[m])P_S[m]\delta_m}{2}\right), \quad \forall m, \tag{10b}$$

$$\mathcal{E}_R \geq \sum_{m=1}^{M}\left(\left(\frac{(1-\alpha[m])P_R[m]}{2}\right)\delta_m\right), \quad \forall m, \tag{10c}$$

$$\mathcal{E}_J \geq \sum_{m=1}^{M}\left(\left((1-\alpha[m])P_J[m]\right)\delta_m\right), \quad \forall m, \tag{10d}$$

$$\frac{1}{M}\sum_{m=1}^{M} P_S[m] \leq P_S^{\text{avg}}[m], \quad \forall m, \tag{10e}$$

$$\frac{1}{M}\sum_{m=1}^{M}(P_i[m]+P_{i,f}[m]) \leq \beta_i^{\max}, \quad \forall m, i, \tag{10f}$$

$$0 \leq P_S[m] \leq P_{S,\max}[m], \quad \forall m, \tag{10g}$$

$$0 \leq P_i[m] \leq P_{i,\max}[m], \quad \forall m, i, \tag{10h}$$

$$\sum_{m=1}^{M} R_{RD}[m] \leq \sum_{m=1}^{M} R_{SR}[m], \ \forall n \in \{2, \cdots M\}, \ \forall m, \tag{10i}$$

$$\mathbf{q}_i[1] = \mathbf{q}_{i_1}, \quad \mathbf{q}_i[M] = \mathbf{q}_{i_F}, \quad \forall i, \tag{10j}$$

$$x_i^2[m] + y_i^2[m] \leq r_{i,\max}^2, \quad \forall m, \tag{10k}$$

$$\|\mathbf{q}_R[m] - \mathbf{q}_J[m]\|^2 \geq d_{\min}, \quad \forall m, \tag{10l}$$

$$\|\mathbf{q}_{i_F} - \mathbf{q}_{i_1}\| \leq v_{i,\max}[m] \times T, \quad \forall m, i, \tag{10m}$$

$$0 \leq \alpha[m] \leq 1, \quad \forall m, \tag{10n}$$

The challenges in solving (P1) primarily arise from two aspects. The objective function of problem (P1), expressed as $R_{\text{sec}}[m] = [R_{\text{leg}}[m] - \max_{\Delta x_n, \Delta y_n \in \mathbb{E}n} R_{\text{wir}}[m]]^+$, is non-smooth due to the mathematical operator $[\cdot]^+$. Also, (P1) exhibits non-convex characteristics stemming from non-convex constraints and multiple coupled variables. In (10a), both the numerator and denominator of $R_{\text{leg}}[m]$ and $R_{\text{wir}}[m]$ involve tightly coupled binary discrete variables, such as $\alpha[m]$, $P_i[m]$, and $\mathbf{q}_i[m]$. In section IV, we employ BCD and SCA methods to decompose (P1) into five subproblems and provide an efficient iterative algorithm to acquire a feasible solution.

IV. Proposed Iterative Algorithm for ASR Maximization Problem

First, we handle the non-smoothness of objective function in (10a), then tackle the uncertainty constraint of E_n, and followed by solving the maximization problem iteratively. We assume that the legitimate rate is greater than the wiretap one, which makes the term $(R_{\text{leg}}[m] - \max_{\Delta x_n, \Delta y_n \in \mathbb{E}_n} R_{\text{wir}}[m])$ in (10a) always a non-negative function [12], [16], [18], [20], [28]. Therefore, the operator $[\cdot]^+$ can be removed without changing the optimal solution to (P1). However, the inclusion of imperfect eavesdroppers' locations ($\Delta x_n[m]$, $\Delta y_n[m]$) introduces a semi-infinite number of constraints that further complicates the mathematical tractability of the optimization problem. Therefore, we first relax the uncertainty constraint of $\max_{\Delta x_n, \Delta y_n \in \mathbb{E}_n} R_{\text{wir}}[m]$ term present in (9) and transform it into its corresponding upper bound expression. Then, we relax the operator $[\cdot]^+$ and substitute (27) in (9) to acquire the achievable SR's lower bound that is given in Lemma 1.

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***Lemma:* 1.** *The lower bound of ASR and reformulate (P1) as*

$$\textit{(P3)}: \max_{\alpha[m],P_i[m],\boldsymbol{q}_i[m]} \frac{1}{M}\sum_{m=1}^{M} R_{sec}^{lb}[m]$$
$$\textbf{s.t.} \quad (10b)\ldots(10n), \tag{11}$$

where $R_{sec}^{lb}[m] = \big[R_{leg}[m] - R_{wir}^{ub}[m]\big]$.

Proof:. *Please refer Appendix A for the proof.*

Nevertheless, the objective function in (P3) is still non-concave and has several non-convex constraints. With this discussion, we use the BCD method to decompose (P3) into five subproblems to obtain a suboptimal solution. Specifically, we alternately optimize the subproblems of TSR, transmit powers of S, U_R, and U_J, and trajectories of U_R and U_J separately while keeping other variables constant. The proposed suboptimal algorithm operates with a polynomial-time computational complexity, making it well-suited for real-time implementation. In the designed system, the algorithm can be executed either by the UAV's onboard computing system or through computation offloading to a ground station [34].

A. Subproblem 1: Optimization of TSR

Given the feasible transmit powers of S, U_R and U_J, and trajectories of U_R and U_J. Consequently, the constraints (10g)-(10m) are relaxed for solving subproblem 1. Further, we relax the binary variables in (10n) into continuous variables [33], [39], [40] to make it mathematically tractable. However, the objective function in (P3) consists of $\min(\cdot,\cdot)$ operator, which makes it difficult to solve. Therefore, we introduce a slack variable $S_{v1}[m]$ to transform the objective function in concave form and formulate the TSR optimization subproblem as

$$\text{(P4)}: \max_{\alpha[m],S_{v1}[m]} \frac{1}{M}\sum_{m=1}^{M} S_{v1}[m] \tag{12a}$$
$$\textbf{s.t.} \quad S_{v1}[m] \le \frac{(1-\alpha[m])\delta_m}{2} A_1[m], \tag{12b}$$
$$S_{v1}[m] \le \frac{(1-\alpha[m])\delta_m}{2} A_2[m], \tag{12c}$$
$$0 \le \sum_{m=1}^{M}\left(A_3[m]\alpha[m] - \frac{(1-\alpha[m])P_S[m]}{2}\right)\delta_m, \tag{12d}$$
$$0 \le \sum_{m=1}^{M}\left(A_4[m]\alpha[m] - \frac{(1-\alpha[m])P_R[m]}{2}\right)\delta_m, \tag{12e}$$
$$0 \le \sum_{m=1}^{M}\left(A_5[m]\alpha[m] - (1-\alpha[m])P_J[m]\right)\delta_m, \tag{12f}$$
$$0 \le \alpha[m] \le 1, \tag{12g}$$

where $A_1[m] = (\log_2(1 + a_1[m]P_S[m]) - A_6[m])$, $A_2[m] = (\log_2(1 + a_2[m]P_R[m]) - A_6[m])$, $A_3[m] = \eta P_B[m]g_{BS}$, $A_4[m] = \eta P_B[m]g_{BR}$, $A_5[m] = \eta P_B[m]g_{BJ}$, $A_6[m] = \log_2(1 + \sum_{n=1}^{N}(a_3[m]))$, $a_1[m] = g_{SR}[m]/\sigma_R^2$, $a_2[m] = g_{RD}[m]/\sigma_D^2$, $a_3 = \tilde{g}_{SE_n}/\sigma_{E_n}^2$, $a_{31}[m] = \tilde{g}_{JE_n}[m]/\sigma_{E_n}^2$, $a_{32}[m] = \tilde{g}_{RE_n}[m]/\sigma_{E_n}^2$, $a_3[m] = \left(\frac{a_{31}P_S[m]+a_{32}[m]P_R[m]}{a_{31}[m]P_J[m]+1}\right)$. In (P4), constraints (12b) and (12c) are incurred due to the introduction of slack variable $S_{v1}[m]$, whereas (12d), (12e), and (12f), are obtained by simplifying (10b), (10c), and (10d). So, (P4) is converted into a standard linear programming problem, which is solved by using the simplex method [33].

B. Subproblem 2: Optimization of Transmit Power of S and U_R

In this section, we optimize the transmit powers of S and U_R for the fixed $\alpha[m]$, $P_J[m]$ and $\mathbf{q}_i[m]$. Consequently, the constraints (10d), (10j)-(10n) are relaxed for solving subproblem 2. Due to the objective function and non-convex constraint (10i), (P3) is a non-convex optimization problem. So, we introduce a slack variable $S_{v2}[m]$, to transform the objective function into a concave form. However, the constraints are still non-convex. Therefore, using the SCA method, we derive an approximate subproblem corresponding to (P3) in Lemma 2.

***Lemma:* 2.** *The convex form of subproblem 2 is given as*

$$\textit{(P5.1)}: \max_{P_S[m],P_R[m],S_{v2}[m]} \frac{1}{M}\sum_{m=1}^{M} S_{v2}[m] \tag{13a}$$
$$\textbf{s.t.} \quad S_{v2}[m] \le \alpha_f[m]\Big(\log_2(1 + a_1[m]P_S[m]) - \sum_{n=1}^{N}\Big(B_1^{\psi}[m]P_S[m] - B_2^{\psi}[m]P_R[m]\Big) + B_3^{\psi}[m]\Big), \tag{13b}$$
$$S_{v2}[m] \le \alpha_f[m]\Big(\log_2(1 + a_2[m]P_R[m]) - \sum_{n=1}^{N}\Big(B_1^{\psi}[m]P_S[m] - B_2^{\psi}[m]P_R[m]\Big) + B_3^{\psi}[m]\Big), \tag{13c}$$
$$0 \le \mathcal{E}_S - \sum_{m=1}^{M}\left(\alpha_f[m]P_S[m]\right), \tag{13d}$$
$$0 \le \mathcal{E}_R - \sum_{m=1}^{M}\left(\alpha_f[m]P_R[m]\delta_m\right), \tag{13e}$$
$$\sum_{m=1}^{M} P_R[m] \le M\beta_R^{\max} - \sum_{m=1}^{M} P_{R,f}[m], \tag{13f}$$
$$\sum_{m=1}^{M}\Big(\log_2\left(1 + a_1[m]P_S[m]\right) - B_4^{\psi}P_R[m] + B_5^{\psi}[m]\Big) \le 0, \tag{13g}$$
$$\textit{(10e), (10g), (10h)},$$

where $B_4^{\psi}[m] = b_1[m]/(1 + b_1[m]P_R^{\psi}[m])$ *and* $B_5^{\psi}[m] = \frac{b_1[m]P_R^{\psi}[m]}{(1+b_1[m]P_R^{\psi}[m])} - log_2(1+b_1[m]P_R^{\psi}[m])$.

Proof:. *The proof of Lemma 2 is provided in Appendix B*

In (P5.1), constraints (13b) and (13c) are incurred due to the introduction of slack variable $S_{v2}[m]$, whereas (13d), (13e), (13f), and (13g) are obtained by simplifying (10b), (10c), (10f), and (10i). Furthermore, the constraints (10g), and (10h) can be strictly followed by taking $P_S[m] \ge 0$ and $P_R[m] \ge 0$, for satisfying (10f) the value of $\beta_R^{\max}$ must be taken high. Now, (P5.1) is transformed into a convex optimization problem and can be reformulated into dual problem to obtain the optimal solution. So, Lagrangian function associated with (P5.1) is given in (15), where $\lambda[m]$ represents $\lambda_1[m], \lambda_2[m], \lambda_3[m], \lambda_4[m], \lambda_5[m], \lambda_6[m]$. So, the Lagrange dual function for the given problem (P5.1) is given by $g(\lambda) = \max_{P_S[m],P_R[m]} \mathcal{L}(P_S[m], P_R[m], \lambda[m])$. Now, we can write the stationarity conditions for $P_S[m]$, and $P_R[m]$

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as $\frac{\partial(\mathcal{L}(P_S[m],P_R[m],\lambda[m]))}{\partial P_S[m]}=0$, $\frac{\partial(\mathcal{L}(P_S[m],P_R[m],\lambda[m]))}{\partial P_R[m]}=0$. The primal conditions for Karush Kuhn Tucker are given as

$$S_{v2}[m]-\alpha_f[m]\Big(\log_2(1+a_1[m]P_S[m]) - \sum_{n=1}^{N}\Big(B_1^{\psi}[m]P_S[m]-B_2^{\psi}[m]P_R[m]\Big)+B_3^{\psi}[m]\Big)\leq 0, \quad (14a)$$

$$S_{v2}[m]-\alpha_f[m]\Big(\log_2(1+b_1[m]P_R[m]) - \sum_{n=1}^{N}\Big(B_1^{\psi}[m]P_S[m]-B_2^{\psi}[m]P_R[m]\Big)+B_3^{\psi}[m]\Big)\leq 0, \quad (14b)$$

$$\mathcal{E}_S-\sum_{m=1}^{M}(\alpha_f[m]P_S[m])\geq 0, \quad (14c)$$

$$\mathcal{E}_R-\sum_{m=1}^{M}(\alpha_f[m]P_R[m]\delta_m)\geq 0, \quad (14d)$$

$$\sum_{m=1}^{M}P_R[m]\leq M\beta_R^{\max}-\sum_{m=1}^{M}P_{R,f}[m], \quad (14e)$$

$$\sum_{m=1}^{M}\Big(\log_2(1+a_1[m]P_S[m])-B_4^{\psi}P_R[m]+B_5^{\psi}[m]\Big)\leq 0, \quad (14f)$$

The dual feasibility conditions for the Lagrangian multipliers is given as $\lambda[m]\geq 0$. Followed by the complementary slackness conditions, which are given as

$$\lambda_1[m](S_{v2}[m]-\alpha_f[m]\Big(\log_2(1+a_1[m]P_S[m]) -\sum_{n=1}^{N}\Big(B_1^{\psi}[m]P_S[m]-B_2^{\psi}[m]P_R[m]\Big)+B_3^{\psi}[m])\Big)=0, \quad (19a)$$

$$\lambda_2[m](S_{v2}[m]-\alpha_f[m]\Big(\log_2(1+b_1[m]P_R[m]) -\sum_{n=1}^{N}\Big(B_1^{\psi}[m]P_S[m]-B_2^{\psi}[m]P_R[m]\Big)+B_3^{\psi}[m])\Big)=0, \quad (19b)$$

$$\lambda_3[m]\left(\mathcal{E}_S-\sum_{m=1}^{M}(\alpha_f[m]P_S[m])\right)=0, \quad (19c)$$

$$\lambda_4[m]\left(\mathcal{E}_R-\sum_{m=1}^{M}(\alpha_f[m]P_R[m]\delta_m)\right)=0, \quad (19d)$$

$$\lambda_5[m]\left(\sum_{m=1}^{M}P_R[m]-M\beta_R^{\max}+\sum_{m=1}^{M}P_{R,f}[m]\right)=0, \quad (19e)$$

$$\lambda_6[m]\sum_{m=1}^{M}\Big(\log_2(1+a_1[m]P_S[m])-B_4^{\psi}P_R[m]+B_5^{\psi}[m]\Big)=0. \quad (19f)$$

After solving these equations, we obtain the optimal solutions, i.e., $P_S^{\text{opt}}[m]$ and $P_R^{\text{opt}}[m]$, which are given by (16) and (17), where $B_6^{\psi}=M\beta_R^{\max}-\sum_{m=1}^{M}P_{R,f}[m]$. Subsequently, subgradient based method is employed to solve the dual problem [31], shown in Algorithm 1, and the subgradient at $[\lambda_1[m],\lambda_2[m],\lambda_3[m],\lambda_4[m],\lambda_5[m],\lambda_6[m]]$ is given by d_{sg} expressed in (18).

C. Subproblem 3: Optimization of U_J's Transmit Power

In this section, we optimize U_J's transmit power by keeping other variables constant. So, the constraints (10b), (10c), (10e), (10i)-(10n) are relaxed for solving subproblem 3. Similar to previous subproblems, we substitute the slack variable, $S_{v3}[m]$, in the subproblem for addressing the non-convex nature of the objective function. Then, we employ the SCA method to acquire a global underestimate of the term as $-\log_2(1+\sum_{n=1}^{N}(\frac{C_1[m]}{P_J[m]c_{11}[m]+1}))\geq\sum_{n=1}^{N}(C_2^{\psi}[m]P_J[m]-P_J^{\psi}[m])+C_3^{\psi}[m]$, where P_J^{ψ} is U_J's transmit power at ψ_{th} iteration, $C_1[m]=\left(\frac{\tilde{g}_{SE_n}[m]}{\tilde{g}_{JE_n}[m]}P_S[m]+\frac{\tilde{g}_{RE_n}[m]}{\tilde{g}_{JE_n}[m]}P_R[m]\right)c_{11}[m]$, $c_{11}[m]=a_{31}^{-1}[m]$, $C_2^{\psi}[m]=\frac{-C_1[m]c_{11}[m]}{\log_2\left(c_{11}[m]P_J^{\psi}[m]+1\right)\left(c_{11}[m]P_J^{\psi}[m]+C_1[m]+1\right)}$, $C_3^{\psi}[m]=-\log_2\left(1+\sum_{n=1}^{N}(\frac{C_1[m]}{P_J[m]+C_{11}[m]})\right)$. Therefore, the subproblem of optimization of U_J's transmit power is formulated as

Algorithm 1: Subgradient Descent Method

Data: Objective function $\mathcal{L}$, step size β, $P_S^0[m]$, $P_R^0[m]$, $\lambda[m]$, maximum iterations ($\Psi^m=100$), tolerance (ϵ)
Result: Optimal solutions $P_S[m]$, $P_R[m]$, $\lambda[m]$, objective value $\mathcal{L}(P_S[m],P_R[m],\lambda[m])$
Initialization:
Initialize $P_S[m]$, $P_R[m]$, $\lambda[m]$;
Select step size (Ω), Ψ^m and ϵ;
for *iteration* = 1 *to maximum iterations* **do**
  Compute $\frac{\partial\mathcal{L}}{\partial P_S[m]}$, $\frac{\partial\mathcal{L}}{\partial P_R[m]}$, $\frac{\partial\mathcal{L}}{\partial\lambda[m]}$
  Parameter Update:
  Update $P_S[m]$: $P_S[m]\leftarrow P_S[m]+\Omega\frac{\partial\mathcal{L}}{\partial P_S[m]}$;
  Update $P_R[m]$: $P_R[m]\leftarrow P_R[m]+\Omega\frac{\partial\mathcal{L}}{\partial P_R[m]}$;
  Update $\lambda[m]$: $\lambda[m]\leftarrow\lambda[m]+\Omega\frac{\partial\mathcal{L}}{\partial\lambda[m]}$;
  Bounds Handling:
  Check and set parameters within bounds;
  if ($\|\nabla\mathcal{L}\|<\epsilon$), **break**;
  Update Best Values and optimal values, **Repeat**
end
Output: Optimal solutions $P_S[m]$, $P_R[m]$, $\lambda[m]$, $\mathcal{L}(P_S[m],P_R[m],\lambda[m])$.

$$\text{(P6)}: \max_{P_J[m]} \frac{1}{M}\sum_{m=1}^{M}S_{v3}[m] \quad (20a)$$

$$\text{s.t.} \quad S_{v3}[m]\leq\alpha_f[m]\Big(\log_2(1+a_1[m]P_S[m]) + \sum_{n=1}^{N}\Big(C_2^{\psi}[m]P_J[m]-P_J^{\psi}[m]\Big)+C_3^{\psi}[m]\Big), \quad (20b)$$

$$S_{v2}[m]\leq\alpha_f[m]\Big(\log_2(1+a_2[m]P_R[m]) + \sum_{n=1}^{N}\Big(C_2^{\psi}[m]P_J[m]-P_J^{\psi}[m]\Big)+C_3^{\psi}[m]\Big), \quad (20c)$$

(10d), (10f), (10h).

It is evident that subproblem (P6) is transformed into a convex optimization problem, which can be effectively solved using a MATLAB convex optimization tool, i.e., CVX [8], [9], [13], [15], [30], [33].

D. Subproblem 4: Optimization of the Trajectory of U_R

Now, we optimize the trajectory of U_R by keeping the other variables constant. Consequently, the constraints (10b)-(10i), (10n) are relaxed for solving subproblem 4. Similar to previous subproblems, we introduce a slack variable, $S_{v4}[m]$, to tackle the $\min[\cdot,\cdot]$ in the objective function. As some of the

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$$\begin{aligned}\mathcal{L}(P_S[m],P_R[m],\lambda[m]) = &\frac{1}{M}\sum_{m=1}^{M} S_{v2}[m] - \sum_{m=1}^{M}\lambda_1[m]\left(S_{v2}[m]-\alpha_f[m]\left(\log_2(1+a_1[m]P_S[m]) - \sum_{m=1}^{M}(C_1^{\psi}[m]P_S[m]-D_1^{\psi}[m]P_R[m])F_1^{\psi}[m]\right)\right)\\ &- \sum_{m=1}^{M}\lambda_2[m](S_{v2}[m]-\alpha_f[m](\log_2(1+b_1[m]P_R[m]) - \sum_{m=1}^{M}(C_1^{\psi}[m]P_S[m]-D_1^{\psi}[m]P_R[m]) + F_1^{\psi}[m]))\\ &- \sum_{m=1}^{M}\lambda_3[m]\left(\mathcal{E}_S - \sum_{m=1}^{M}(\alpha_f[m]P_S[m])\right) - \sum_{m=1}^{M}\lambda_4[m]\times\left(\mathcal{E}_R - \sum_{m=1}^{M}(\alpha_f[m]P_R[m]\delta_m)\right)\\ &- \sum_{m=1}^{M}\lambda_5[m]\left(\sum_{m=1}^{M}P_R[m] - M\beta_R^{\max} + \sum_{m=1}^{M}P_{R,f}[m]\right)\\ &- \sum_{m=1}^{M}\lambda_6[m]\left(\sum_{m=1}^{M}\left(\log_2(1+a_1[m]P_S[m]) - B_4^{\psi}P_R[m] + B_5^{\psi}[m]\right)\right),\end{aligned} \tag{15}$$

$$P_S^{\text{opt}}[m] = \max\left(\min\left(\frac{(\lambda_1[m]+\lambda_2[m])\left(\sum_{n=1}^{N}B_1^{\psi}[m]\right) + \frac{\lambda_3[m]\alpha_f[m]}{\ln 2} - a_1[m](\lambda_1[m]\alpha_f[m] - \frac{\lambda_6[m]}{\ln 2}) + \lambda_5[m]B_6}{\left(\frac{\lambda_3[m]\alpha_f[m]}{\ln 2} + (\lambda_1[m]+\lambda_2[m])\left(\sum_{n=1}^{N}B_1^{\psi}[m]\right)\right)a_1[m]},\ P_{S,\max}\right),0\right), \tag{16}$$

$$P_R^{\text{opt}}[m] = \max\left(\min\left(\frac{\lambda_2[m]\alpha_f[m]b_1[m] - \frac{\lambda_4[m]\alpha_f[m]}{\ln 2} - \frac{\lambda_5[m]B_4^{\psi}[m]}{\ln 2} - (\lambda_1[m]+\lambda_2[m])\left(\sum_{n=1}^{N}B_2^{\psi}[m]\right) + \lambda_5[m]B_6}{\left(\frac{\lambda_4[m]\alpha_f[m]}{\ln 2} - \frac{\lambda_5[m]B_4^{\psi}[m]}{\ln 2} - (\lambda_1[m]+\lambda_2[m])\left(\sum_{n=1}^{N}B_2^{\psi}[m]\right)\right)a_2[m]}, P_{R,\max}\right),0\right), \tag{17}$$

$$\begin{aligned}d_{sg} = \Bigg[&\alpha_f[m](\log_2(1+a_1[m]P_S^{\text{opt}}[m])),\quad \alpha_f[m](\log_2(1+a_2[m]P_R^{\text{opt}}[m])),\quad \sum_{m=1}^{M}\alpha_f[m]P_S^{\text{opt}}[m],\\ &\sum_{m=1}^{M}\alpha_f[m]P_R^{\text{opt}}[m],\quad \sum_{m=1}^{M}P_R^{\text{opt}}[m],\quad \sum_{m=1}^{M}\log_2(1+a_1[m]P_S^{\text{opt}}[m])\Bigg].\end{aligned} \tag{18}$$

constraints in the reformulated problem are non-convex, the optimization problem is challenging to solve. So, we introduce auxiliary variables $\mathbb{L}_{SR}[m] = \{L_{SR}[1],\ldots,L_{SR}[m]\}$, $\mathbb{M}_{RE}[m] = \{M_{RE}[1],\ldots,M_{RE}[m]\}$, $\mathbb{N}_{RD}[m] = \{N_{RD}[1],\ldots,N_{RD}[m]\}$ in the constraints to perform convex processing. It should be noted from Appendix C that the equality in constraint (30d) must be satisfied at the optimal solution. Otherwise, we could increase the objective function value without violating any constraints by decreasing $\mathbb{L}_{SR}[m]$. Similarly, the same reasoning applies to constraints (30e) and (30f). Therefore, problems (P7) and the original problem are equivalent and pose the same optimal solution. Moreover, $D_1[m] = \rho_0 P_S[m]/\sigma_R^2$, $D_3[m] = \rho_0 P_R[m]/\sigma_D^2$, and $D_2 = \rho_0 P_R[m]/c_1[m]P_S[m]\sigma^2(c_2[m]P_J[m]+1)$. Then, we transform the non-convex constraints into the convex ones and reformulate the U_R's trajectory problem in Lemma 3.

***Lemma:* 3.** *The subproblem of optimization of U_R trajectory is transformed as*

$$\textit{(P7.1)}: \max_{\substack{\boldsymbol{q}_R[m],S_{v4}[m]\\ \mathbb{L}_{SR}[m],\mathbb{M}_{RE}[m],\mathbb{N}_{RD}[m]}} \frac{1}{M}\sum_{m=1}^{M}\alpha_f[m]S_{v4}[m] \tag{21a}$$

$$\textit{s.t.}\ S_{v4}[m] \le \left(D_4^{\psi}[m]\mathbb{L}_{SR}[m] + D_5^{\psi}[m] - \log_2\left(1+\sum_{n=1}^{N}\frac{D_2[m]}{\mathbb{M}_{RE}}\right)\right), \tag{21b}$$

$$S_{v4}[m] \le \left(D_6^{\psi}[m]\mathbb{N}_{RD}[m] + D_7^{\psi}[m] - \log_2\left(1+\sum_{n=1}^{N}\frac{D_2[m]}{\mathbb{M}_{RE}}\right)\right), \tag{21c}$$

$$\mathbb{L}_{SR}[m] \le f_1(\boldsymbol{q}_R[m],\boldsymbol{q}_R^{\psi}[m]) + H_R^2, \tag{21d}$$

$$\mathbb{M}_{RE}[m] \ge (\|\boldsymbol{q}_R[m] - \boldsymbol{w}_{E_n}\| - \Delta Q_n)^2 + H_R^2, \tag{21e}$$

$$\mathbb{N}_{RD}[m] \le f_2(\boldsymbol{q}_R[m],\boldsymbol{q}_R^{\psi}[m]) + H_R^2, \tag{21f}$$

$$\sum_{m=1}^{M}\alpha_f[m](D_4^{\psi}[m]\mathbb{L}_{SR}[m] + D_5^{\psi}[m] - D_6^{\psi}[m]\mathbb{N}_{RD}[m] - D_7^{\psi}[m]) \le 0, \tag{21g}$$

$$f_3(\boldsymbol{q}_R[m],\boldsymbol{q}_R^{\psi}[m]) \ge d_{\min}, \tag{21h}$$

(10f), (10j), ..., (10k), (10m).

Proof:. *The proof of Lemma 3 is given in Appendix C.*

Subproblem (P7.1) is transformed into a convex optimization problem which can be effectively solved by using CVX.

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E. Subproblem 5: Optimization of Trajectory of U_J

In this section, we optimize the trajectory of U_J for the fixed TSR, transmit powers of S, U_R, and U_J and the trajectory of U_R. Consequently, the constraints (10b)-(10i), (10n) are relaxed for solving subproblem 4. Similar to previous subproblems, we introduce a slack variable, $S_{v5}[m]$, to tackle the $\min[\cdot,\cdot]$ in the objective function. Therefore, we convert non-convex constraints into convex ones and reformulate U_J's trajectory problem in Lemma 4.

***Lemma:* 4.** *We reformulate the subproblem of optimization of U_J's trajectory as*

$$\textit{(P8.1)}: \max_{\boldsymbol{q}_J[m], S_{v5}[m], \mathbb{O}_{JE}[m]} \frac{1}{M}\sum_{m=1}^{M} \alpha_f[m] S_{v5}[m] \tag{22a}$$

$$\textit{s.t.}\quad S_{v5}[m] \le \log_2\left(1+G_1[m]\right) - G_6^{\psi}[m] - \mathbb{O}_{JE}[m], \tag{22b}$$

$$S_{v5}[m] \le \log_2\left(1+G_2[m]\right) - G_6^{\psi}[m] - \mathbb{O}_{JE}[m], \tag{22c}$$

$$\mathbb{O}_{JE}[m] \ge \left(\|\boldsymbol{q}_R[m] - \boldsymbol{w}_{E_n}\| + \Delta Q_n\right)^2 + H_J^2, \tag{22d}$$

$$f_4(\boldsymbol{q}_J[m], \boldsymbol{q}_J^{\psi}[m]) \ge d_{\min}, \tag{22e}$$

(10f), (10j), ..., (10k), (10m).

Proof:. *The proof of Lemma 4 is given in Appendix D.*

So, subproblem (P8.1) is transformed into a convex optimization problem which can be effectively solved by using CVX [23].

F. Proposed Algorithm and Computational Complexity

We efficiently obtain solutions for the five subproblems in the previous sections III-A, B, C, D, and E. Herein, we propose an iterative algorithm combining the solutions of these subproblems based on the BCD method to solve the original problem (P3). Algorithm 2 alternately optimizes wireless charging duration $\alpha[m]$, transmit powers $P_i[m]$, and trajectories $\mathbf{q}_i[m]$ separately, while keeping other variables constant. Let $F(\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi}) = \frac{1}{M}\sum_{m=1}^{M} R_{\text{sec}}^{\text{lb}}$, where $P_{\zeta}^{\psi} \in \{P_S^{\psi}, P_R^{\psi}\}$ represent the objective function of the original problem (P3) at the ψth iteration.

The initialization process of the algorithm begins by setting $\psi = 0$ with maximum iterations $\psi^m = 100$. The transmit powers are initialized to their average values, i.e., $P_{\zeta}^0 = P_{\zeta}^{\text{avg}} = 20$ dBm and $P_J^0 = P_J^{\text{avg}} = 20$ dBm, $P_{\zeta}^{\text{avg}} = 20$ dBm. The initial positions of the UAVs are set to their starting coordinates, i.e., $\mathbf{q}_R^0 = \mathbf{q}_{R1} = (0, -150)$ for the relay UAV and $\mathbf{q}_J^0 = \mathbf{q}_{J1} = (0, -50)$ for the jammer UAV. These positions serve as the initial locations from which the UAVs will begin their trajectories. Slack variables and the auxiliary variables are initialized to their default values. The objective function $F(\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi})$ is initialized to 0, establishing a baseline for evaluating progress during iterations. The solutions acquired in each iteration are utilized as feasible input points for the subsequent iteration. At each step, the objective function $F(\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi})$ is updated until the change in F between iterations satisfies the convergence threshold μ, ensuring an efficient and practical optimization process. The selection of initialization parameters is essential to prevent suboptimal solutions and ensure faster convergence during the iterative optimization of variables. The specifics of the iterative process are outlined in Algorithm 2.

Algorithm 2: Iterative Algorithm for Problem (P1)

Input: $F(\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi})$, $\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi}$, P_B, $\beta_i^{\max}$, $\mu = 0.001$, $T = 80$ sec $\psi^m = 100$;

Initialization: Set the initial feasible points: $\psi = 0$, $\alpha^0 = \alpha^0[m]$, Transmit powers are initialized to the average powers, i.e., $P_{\zeta}^0 = P_{\zeta}^0[m] = P_{\zeta}^{\text{avg}}$, $P_J^0 = P_J^0[m] = P_J^{\text{avg}}$.
Set the UAVs' initial position: $\mathbf{q}_R^0 = \mathbf{q}_R^0[m] = \mathbf{q}_{R1}$, $\mathbf{q}_J^0 = \mathbf{q}_J^0[m] = \mathbf{q}_{J1}$.
Set the initial values of slack variables: $S_{v1}[m]$, $S_{v2}[m]$, $B_1^{\psi}[m]$, $B_2^{\psi}[m]$, $B_3^{\psi}[m]$, $B_4^{\psi}[m]$, $B_5^{\psi}[m]$, $S_{v3}[m]$, $C_2^{\psi}[m]$, $C_3^{\psi}[m]$, $S_{v4}[m]$, $L_{SR}^{\psi}[m]$, $N_{RD}^{\psi}[m]$, $D_4^{\psi}[m]$, $D_5^{\psi}[m]$, $D_6^{\psi}[m]$, $D_7^{\psi}[m]$, $S_{v5}[m]$, $O_{JE}^{\psi}[m]$, $G_6^{\psi}[m]$.
Set $F(\alpha^0, P_{\zeta}^0, P_J^0, \mathbf{q}_R^0, \mathbf{q}_J^0) = 0$.

while $F(\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi}) - F(\alpha^{\psi-1}, P_{\zeta}^{\psi-1}, P_J^{\psi-1}, \mathbf{q}_R^{\psi-1}, \mathbf{q}_J^{\psi-1}) \le \mu$ **do**

1) Solve (P4) for given $\{(P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi})\}$ and obtain the solution $\alpha^{\psi+1}$.
2) Solve (P5.1) for given $\{(\alpha^{\psi+1}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi})\}$ and obtain the solution $P_{\zeta}^{\psi+1}$. Update $S_{v2}[m]$, $B_1^{\psi}[m]$, $B_2^{\psi}[m]$, $B_3^{\psi}[m]$, $B_4^{\psi}[m]$, $B_5^{\psi}[m]$.
3) Solve (P6) for given $\{(\alpha^{\psi+1}, P_{\zeta}^{\psi+1}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi})\}$ and obtain the solution $P_J^{\psi+1}$. Update $S_{v3}[m]$, $C_2^{\psi}[m]$, $C_3^{\psi}[m]$.
4) Solve (P7.1) for given $\{(\alpha^{\psi+1}, P_{\zeta}^{\psi+1}, P_J^{\psi+1}, \mathbf{q}_J^{\psi})\}$ and obtain the solution $\mathbf{q}_R^{\psi+1}$. Update $S_{v4}[m]$, $L_{SR}^{\psi}[m]$, $N_{RD}^{\psi}[m]$, $D_4^{\psi}[m]$, $D_5^{\psi}[m]$, $D_6^{\psi}[m]$, $D_7^{\psi}[m]$.
5) Solve (P8.1) for given $\{(\alpha^{\psi+1}, P_{\zeta}^{\psi+1}, P_J^{\psi+1}, \mathbf{q}_R^{\psi+1})\}$ and obtain the solution $\mathbf{q}_J^{\psi+1}$.
Update $S_{v5}[m]$, $O_{JE}^{\psi}[m]$, $G_6^{\psi}[m]$.
6) Update $\psi \Rightarrow \psi + 1$.
7) Compute the objective value: $F(\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi})$.

end

Output: $F(\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \mathbf{q}_R^{\psi}, \mathbf{q}_J^{\psi})$.

1) Convergence Analysis: The convergence of the proposed Algorithm 2 is provided in Lemma 5.

***Lemma:* 5.** *Algorithm 2 is convergent in behavior.*

Proof:. *We solve a standard LPP in Step 1 of Algorithm 2 and obtain $(\alpha^{\psi+1})$. Consequently, by solving (P5.1), we acquire a suboptimal solution $P_{\zeta}^{\psi+1}$. We have*

$$\begin{aligned} &F(\alpha^{\psi+1}, P_{\zeta}^{\psi}, P_J^{\psi}, \boldsymbol{q}_R^{\psi}, \boldsymbol{q}_J^{\psi}) \\ &\qquad \overset{(i)}{=} F^{lb}(\alpha^{\psi+1}, P_{\zeta}^{\psi}, P_J^{\psi}, \boldsymbol{q}_R^{\psi}, \boldsymbol{q}_J^{\psi}) \\ &\qquad \overset{(ii)}{\le} F^{lb}(\alpha^{\psi+1}, P_{\zeta}^{\psi+1}, P_J^{\psi}, \boldsymbol{q}_R^{\psi}, \boldsymbol{q}_J^{\psi}) \\ &\qquad \overset{(iii)}{\le} F(\alpha^{\psi+1}, P_{\zeta}^{\psi+1}, P_J^{\psi}, \boldsymbol{q}_R^{\psi}, \boldsymbol{q}_J^{\psi}), \end{aligned} \tag{23}$$

where the objective function of the approximated problem (P5.1) is given by $F^{lb}(\alpha^{\psi}, P_{\zeta}^{\psi}, P_J^{\psi}, \boldsymbol{q}_R^{\psi}, \boldsymbol{q}_J^{\psi})$. To satisfy (23)*, the following conditions must be satisfied: (i) ensuring that lower bound to objective function in (P3) is given by (P5.1) and* (23) *illustrates that ASR consistently increases after each iteration; (ii) considering the first-order Taylor expansion's accuracy at a feasible point; and (iii) following the optimality of solving (P5.1) [10], [16], [18], [20]. It is proved that the objective of the problem (P3) is a non-decreasing function after each iteration. However, the maximum ASR is limited by a finite value due to the imposed constraints. Therefore, Algorithm 2 converges to a local suboptimal solution for problem (P3).*

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2) Computational Complexity Analysis: The overall computational complexity of Algorithm 2 is determined via complexity of its iterative steps. The complexity of solving (P4) in Step 1 is $O[M^{3.5}\log(1/\mu)]$, where μ is the given tolerance level [33], [41]. Solving (P5.1) in Step 2 has a complexity of $O[M^2\nu^{0.5}]$, where ν is the number of updates for feasible initial points $P_S^0[m]$ and $P_R^0[m]$. For Steps 3, 4, and 5, which involve solving (P6), (P7.1), and (P8.1), the complexity of each step is $O[(5M)^{3.5}\log(1/\mu)]$. The total computational complexity for one iteration of Algorithm 2 is $O[(5M)^{3.5}\log(1/\mu)]$. Considering the number of iterations required for convergence (ψ), overall complexity of Algorithm 2 is given as [5]

$$\mathcal{O}\left(\left(M^{3.5}\log\left(\frac{1}{\mu}\right)+M^2\nu^{0.5}+(5M)^{3.5}\log\left(\frac{1}{\mu}\right)\right)\psi\right). \quad (24)$$

The computational complexity of Algorithm 2 is higher compared to the methods in [13], [16]–[18], [20], [28], due to the additional optimization block in the BCD framework. The authors in [28] optimized $\alpha[m]$, power and trajectory of a UAV for only A2G link. Similarly, the approaches in [13], [16]–[18], [20] only focused on optimizing the trajectories and transmit powers of U_R and U_J. In contrast, our proposed algorithm jointly optimizes $\alpha[m]$, $\mathbf{q}_i[m]$, and $P_i[m]$, which increases the complexity but provides a more comprehensive solution to the problem. The iterations required for convergence can be reduced by carefully selecting initialization parameters, including μ, T, UAVs' initial positions, and UAVs' and IoT devices' transmit and jamming powers. Thereby lowering the overall computational cost of the proposed algorithm.

TABLE III: Simulation parameters and their values.

Parameter	Value	Parameter	Value
T [5], [29]	120 sec	δ_m [34]	0.5 sec
$\beta_i^{\max}$ [16], [20]	400 W	$\mathbf{q}_{R1}$	$(0,-150)$ m
$\mathbf{q}_{RF}$	$(525,-150)$ m	$\mathbf{q}_{J1}$	$(0,-50)$ m
$\mathbf{q}_{JF}$	$(525,-50)$ m	B	$(0,0)$ m
S	$(100,-30)$ m	D	$(430,-30)$ m
(x_{E_1},y_{E_1})	$(280,120)$ m	(x_{E_2},y_{E_2})	$(290,110)$ m
(x_{E_3},y_{E_3})	$(300,120)$ m	ΔQ_n	30 m
H_i [36], [37]	120 m	$r_{i,\max}$	250 m
$v_{i,\max}$	5 m/sec	$d_{\min}$	20 m
ρ_0 [5], [28]	50 dBm	$\sigma^2,\sigma_R^2,\sigma_D^2,$ $\sigma_{E_n}^2$ [29]	-50 dBm
η	0.8	P_S^{avg} [21]	20 dBm
P_i^{avg} [21]	20 dBm	$P_{S,\max}$ [21]	27 dBm
$P_{i,\max}$ [21]	27 dBm	ϵ, μ	0.005, 0.001

V. Simulation Results

In this section, we present numerical and simulation results to validate the analytical framework and gain valuable insights into the system behavior. Moreover, we show that data secrecy can be significantly improved by designing the trajectories of U_R and U_J and efficient resource allocation in terms of TSR, transmit powers of S, U_R, and U_J. The proposed algorithm jointly optimizes $\alpha[m]$, $\mathbf{q}_i[m]$, and $P_i[m]$, which we refer as α-TRJ-PIRJ. To show the effectiveness of the considered framework, we compare the system performance in terms of achievable ASR with four benchmark schemes. For these schemes, the optimization parameters are categorized as

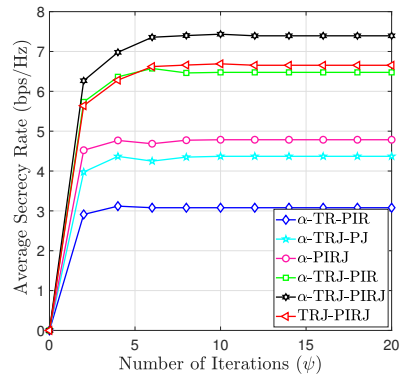

Fig. 2: Convergence analysis of different algorithms.

1) Optimizing $\alpha[m]$, trajectory of U_R with the transmit powers of S and U_R with no U_J (α-TR-PIR) [28].
2) Optimizing $\alpha[m]$, trajectories of U_R and U_J with transmit powers of S and U_R without taking transmit power of U_J into consideration (α-TRJ-PIR).
3) Optimizing $\alpha[m]$, trajectories of U_R and U_J with transmit power of U_J (α-TRJ-PJ). Here, the transmit powers of S and U_R are not considered for optimization.
4) Optimizing $\alpha[m]$, transmit powers of S, U_R, and U_J (α-PIRJ). Here, the trajectories of U_R and U_J are not considered.
5) Optimizing trajectories of U_R and U_J with transmit powers of S, U_R, and U_J without taking $\alpha[m]$ into consideration (TRJ-PIRJ).

In all these benchmark schemes, we consider the optimization of $\alpha[m]$ and modify the proposed algorithm by changing the other parameters. For example in α-TR-PIR scheme, we set $P_J = 0$ and then solve (P4), (P5.1), and (P7.1) iteratively until the convergence is achieved, where $P_J = 0$ represents that jammer is not active in the system. The authors in [28] have considered the α-TR-PIR scheme for their optimization problem that can be compared with the proposed model. Likewise, for α-TRJ-PIR, we solve (P4), (P5.1), (P7.1), and (P8.1) for jointly optimizing the trajectories of U_R and U_J and transmit power of U_R with fixed transmit power of U_J. For α-TRJ-PJ, we solve (P4), (P6), (P7.1), and (P8.1) to optimize the trajectories of U_R and U_J and transmit power of U_J with fixed transmit power of U_R. At the last, for α-PIRJ, we solve (P4), (P5.1), and (P6) until the convergence to optimize the transmit powers of S, U_R and U_J with fixed trajectories of U_R and U_J. Lastly, we consider the TRJ-PIRJ scheme for which we solve (P5.1), (P6), (P7.1), and (P8.1) for jointly optimizing the trajectories of U_R and U_J and transmit powers of S, U_R and U_J with fixed α value, i.e., $\alpha = 0.5$. The authors in [13], [16]–[20] have employed a similar scheme (but restricted only to the A2G link), so we compare this setup with our proposed framework and demonstrate the results in Fig. 2.

A. Simulation Setup and Convergence Analysis

The simulation parameters are selected in accordance with the network architecture and propagation models defined by 3GPP standards. In particular, the channel gain, noise power

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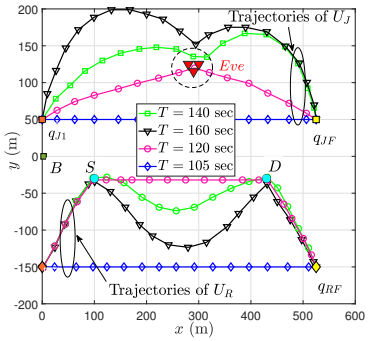

Fig. 3: Trajectories of U_R and U_J for different flight period.

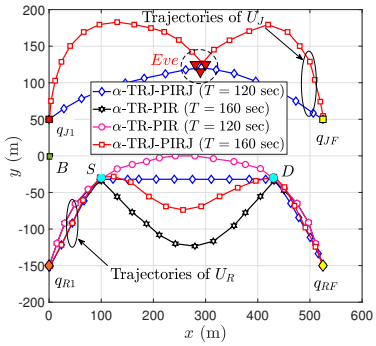

Fig. 4: Trajectories of single UAV scheme and proposed algorithm.

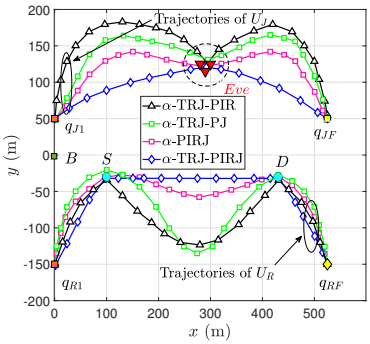

Fig. 5: Trajectories of U_R and U_J for the baseline and proposed algorithm.

levels, and transmit or jamming powers are aligned with 3GPP and ITU-T specifications [11], [16], [21], [29], [33]. The UAVs' height and coverage radius are chosen based on 3GPP guidelines for aerial communication and align with values in the literature [36], [37]. Without any loss of generality, the system parameters are set as in Table III. The convergence behavior of the proposed Algorithm 2 along with the different baseline algorithms are depicted in Fig. 2. As shown in the graph, the achievable ASR of all algorithms initially increases rapidly and then reaches a constant value after several iterations. Specifically, the proposed Algorithm 2 outperforms other baseline schemes by utilizing more computational resources as compared to the other baselines.

B. Trajectories Analysis of U_R and U_J with Different Schemes

In Figs. 3, 4, and 5, we analyze the trajectories of U_R and U_J versus different flight duration for the proposed and baseline schemes. All the involved nodes, i.e., B, S, D, and Eve, are marked in these figures along with the initial and final locations of U_R and U_J. Specifically, Fig. 3 depicts the trajectories of both UAVs for the proposed α-TRJ-PIRJ scheme with different flight periods T. Due to the constraint in (10m), it is evident that U_R and U_J flies straight from their initial points to the final points at a maximum velocity of 5 m/sec for the minimum flight period, i.e., $T = 105$ sec. Therefore, taking $T > 105$ sec can provide better insights into different trajectories of U_R and U_J for our proposed α-TRJ-PIRJ scheme.

The curves in Fig. 3 demonstrate variations in the trajectories of both UAVs for different flight duration. For $T > 105$ sec, the trajectories of U_R and U_J can be categorized into three different stages: (i) Starting from initial locations, U_R and U_J fly to their quasi-stationary areas at maximum velocity, i.e., U_R heads to a position closer to the legitimate nodes to facilitate communication, whereas U_J flies to a location near the Eve; (ii) Upon reaching their respective places, U_R and U_J slow down and maneuver at lower speeds to execute their assigned tasks, i.e., U_R aims to gather confidential information from S and relay it to D using better channel conditions, whereas U_J maneuvers near the Eve to optimize the jamming channel and sends as much jamming signal as possible to confuse it; (iii) After completing their tasks, U_R and U_J fly to their final locations at maximum speed to satisfy the constraint, which guarantees that the UAVs reach their final positions within T. The curves in Fig. 3 show that U_R and U_J follow more curved trajectories as the flight duration increases from $T = 105$ sec to $T = 160$ sec. This indicates that the UAVs can utilize extra time to optimize their paths, resulting in more intricate trajectory optimization.

Fig. 4 shows the trajectories of U_R and U_J for the proposed α-TRJ-PIRJ scheme and U_R for the α-TR-PIR scheme for $T = 120$ sec and $T = 160$ sec. We can observe from the curves that the U_J affects the trajectory of U_R due to the applied constraints. The trajectory of U_R experiences a longer path to avoid eavesdropper during the mission. Fig. 5 shows comparison between the proposed scheme and the α-TRJ-PIR, α-TRJ-PJ, and α-PIRJ schemes. Further, Figs. 4 and 5 demonstrate that the trajectories of U_R and U_J for the benchmark schemes have the same stages as our proposed scheme. However, the trajectory of U_R and U_J benefits from the joint trajectory and power optimization, and U_R and U_J experience shorter flight distance in the proposed α-TRJ-PIRJ scheme compared to another scheme. It also depicts that the trajectories with α-TRJ-PIR and α-TRJ-PJ schemes are similar because U_R and U_J benefit from the trajectory optimization. However, U_J consumes more power for accomplishing the larger trajectories in the α-TRJ-PIR scheme, whereas, for the α-TRJ-PJ scheme, U_R consumes more power for the same.

C. ASR Analysis with Different Parameters

In Figs. 6, 7, 8, 9, and 10, we plot the achievable ASR curves versus different system parameters for the proposed and baseline schemes. Specifically, Fig. 6 shows the ASR versus the transmit power P_B for $T = 120$ sec and $T = 140$ sec. It is observed that the proposed α-TRJ-PIRJ scheme achieves the highest ASR, while the benchmark α-TR-PIR model has the lowest ASR. Additionally, for the lower P_B, the benchmark α-TRJ-PIR scheme achieves an ASR performance close to the proposed α-TRJ-PIRJ and significantly outperforms other benchmark schemes. However, as P_B increases, the ASR of α-PIRJ scheme surpasses that of the α-TRJ-PIR algorithm and even approaches that of the proposed α-TRJ-PIRJ scheme. These findings indicate that there is significant potential for

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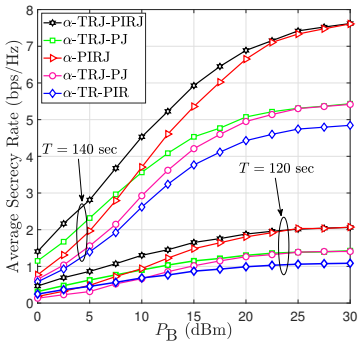

Fig. 6: Achievable ASR of different algorithms versus P_B.

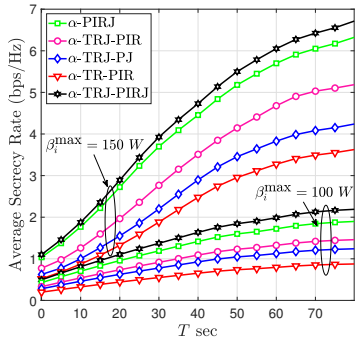

Fig. 7: Achievable ASR versus T for UAVs' limited battery capacity.

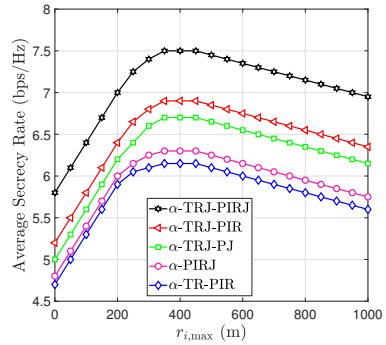

Fig. 8: Achievable ASR of different algorithms versus $r_{\max}$.

improvement in ASR through the optimization of P_B, particularly for lower P_B values. This suggests a strategic focus on optimizing P_B could yield substantial benefits. In contrast, trajectory optimization is more effective when P_B is high.

Fig. 7 illustrates the ASR curves as a function of T with varying UAV battery capacities across different schemes. The proposed algorithm consistently achieves a higher ASR for all values of T. It outperforms the benchmark schemes, showcasing the significance of jointly optimizing UAV trajectory and transmit power control for maximizing secrecy. Among the benchmark schemes, the α-TRJ-PIR demonstrates superior ASR compared to α-TRJ-PJ throughout the entire range of T. This improvement arises because α-TRJ-PIR leverages better channel conditions at U_R to increase the SNR at the legitimate nodes. The single relay scheme shows the worst ASR performance, i.e., α-TR-PIR. Moreover, Fig. 7 reveals an intriguing trend in the ASR performance of all schemes for two different values of $\beta_i^{\max}$, i.e., $\beta_i^{\max} = 360\ W$ and $\beta_i^{\max} = 400\ W$. The ASR performance improves for all schemes as the UAVs' battery capacity increases for the entire region of T. This is because, at lower $\beta_i^{\max}$ values, the power budgets of U_R and U_J are constrained, leading to larger trajectories with greater turning radii. This limitation prevents U_R and U_J from moving closer to their designated nodes, resulting in lower ASR performance. Conversely, when $\beta_i^{\max}$ increases, the additional power budget allows both U_R and U_J to move more quickly toward the ground nodes and eavesdroppers.

Fig. 8 illustrates the achievable ASR curves as a function of the maximum flight radius. It can be observed that the ASR initially increases with the increase in $r_{\max}$ and reaches a peak value. This initial increase is attributed to the improved positioning of U_R and U_J as the coverage radius of the UAVs grows. At smaller values of $r_{i,\max}$, U_R and U_J are restricted in their movement, preventing them from achieving optimal positions to maximize the ASR. As $r_{i,\max}$ increases, U_R can establish a more efficient communication link between S and D, while U_J can position itself to degrade the eavesdroppers' link effectively. This increased flexibility allows for better utilization of the harvested energy, which is governed by the α. However, the ASR declines gradually beyond a certain optimal coverage radius (approximately 300 meters). This behavior arises because, as $r_{i,\max}$ continues to increase, the UAVs eventually reach their optimal positions relative to S, D, and E_n, beyond which further expansion provides no additional ASR gains. Moreover, as $r_{i,\max}$ grows, the distance between B and the UAVs increases. This decreases the power received from B due to path loss, requiring the UAVs to dedicate more time to harvest energy to meet the EH constraints.

Fig. 9 reveals crucial observations about the impact of the achievable ASR for different values of the estimation error radius ΔQ_n and P_B. As ΔQ_n increases, the ASR performance consistently degrades across all considered values of P_B. This is attributed to the heightened uncertainty in the locations of eavesdroppers for larger ΔQ_n, influencing the efficient utilization of system resources and the trajectory design of U_R and U_J. Further, we present the achievable ASR versus different H_i and transmit power P_B for the proposed α-TRJ-PIRJ scheme in Fig. 10. It can be observed that the achievable ASR performance improves with increasing H_i before a certain point for all transmit power P_B values. After that, ASR starts decreasing. This is due to the fact that when H_i is sufficiently large, the distance between UAVs and ground nodes also increases, prompting U_R and U_J to harvest more energy from B by elevating α. Consequently, this leads to a reduction in the time allocated for information transfer. The curves also show that ASR increases with the increase in P_B.

D. Transmit Power Consumption Analysis

Fig. 11 shows the optimized transmit power of U_R and U_J versus varying $T = 120$ sec for the proposed α-TRJ-PIRJ and other benchmark schemes. The curves reveal that S power increases to the peak rapidly, followed by a gradual fall till 0. This occurs because the U_R flies closer to S, which uses more power for transmitting information to U_R with increased throughput due to allocated T. However, as U_R flies away from S, then S decreases the transmit power due to the power constraint. Both figures depict that the transmit power of U_R for the proposed α-TRJ-PIRJ and baseline schemes have a similar trend, i.e., the curve monotonically decreases after attaining a peak power value. At the same time, the power consumption for U_J shows the opposite trend to that of U_R.

The transmit power of U_J is optimized in the proposed α-TRJ-PIRJ and baseline α-TRJ-PJ and α-PIRJ methods. These

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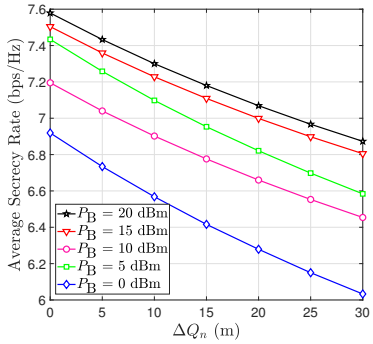

Fig. 9: Achievable ASR with varying ΔQ_n.

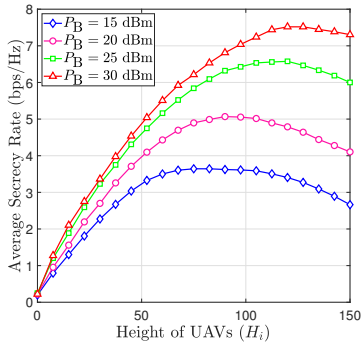

Fig. 10: Achievable ASR with varying heights of U_R and U_J.

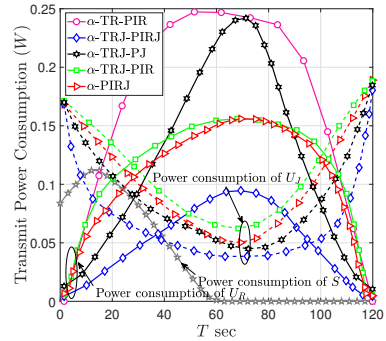

Fig. 11: Transmit power consumption with varying flight period.

curves show a significant improvement in the power consumption of U_J. It is observed that when U_J is far from Eve, it transmits more jamming signals to combat Eve. However, as U_J approaches the Eve, the improved channel condition between them is utilized by the U_J to combat the eavesdropper with relatively lower transmit power. The transmit power curve of the α-TR-PIR model has higher power consumption than other schemes. The variation in power control between the α-TR-PIR and other schemes indicates that including the cooperative U_J enhances resource utilization efficiency.

VI. Conclusion

This paper has considered dual UAV-enabled secure data transfer between an energy-constrained IoT device and a base station in the presence of multiple terrestrial eavesdroppers. We have formulated the ASR maximization problem by considering the joint optimization of wireless charging duration, trajectory, and transmit power for both UAVs. This optimization takes into account various constraints, including limited battery capacity, maximum flying radius, EH, and information causal constraints. To efficiently solve this non-convex problem, we have employed iterative algorithms based on the BCD and SCA optimization methods. Simulation results demonstrated a significant improvement in secrecy performance achieved by the proposed scheme compared to other baseline methods. Compared to other schemes, the proposed α-TRJ-PIRJ scheme experiences shorter trajectories of U_R and U_J due to joint trajectory and power optimization. Furthermore, the transmit power curve of the α-TR-PIR model confirms that incorporating the cooperative U_J improves resource utilization efficiency. While this work has demonstrated the potential of joint optimization in UAV-enabled secure communications, several promising avenues for future research remain unexplored. For instance, integrating machine learning techniques to enable dynamic adaptation to environments with mobile terrestrial nodes can significantly enhance system robustness. Additionally, incorporating a more generalized fading scenario with such systems can offer more detailed insights into the system performance.

Appendix A

Due to the uncertain location of E_n, the legitimate nodes S, U_R, U_J can only estimate E_n's location. By using (4) and (8), we obtain

$$\max_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} R_{\rm wir}[m] = \frac{(1-\alpha[m])\delta_m}{2} \times \log_2\Bigg(1+\sum_{n=1}^{N}\Bigg(\frac{(\max_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} g_{SE_n})P_S[m]}{(\min_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} g_{JE_n}[m])P_J[m]+\sigma_{E_n}^2} + \sum_{n=1}^{N}\Bigg(\frac{(\max_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} g_{RE_n}[m])P_R[m]}{(\min_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} g_{JE_n}[m])P_J[m]+\sigma_{E_n}^2}\Bigg)\Bigg). \tag{25}$$

We employ facts from the literature [18], [42], [43] that states that the optimal channel power gains $\grave{g}_{SE_n}$ and $\grave{g}_{RE_n}[m]$ from S-E_n and U_R-E_n can be obtained at $\mathbf{w}_{SE_n} = \tilde{\mathbf{w}}_{SE_n} + \frac{\mathbf{w}_S-\tilde{\mathbf{w}}_{E_n}}{\|\mathbf{w}_S-\tilde{\mathbf{w}}_{E_n}\|}\Delta Q_n$ and $\mathbf{w}_{RE_n}[m] = \tilde{\mathbf{w}}_{RE_n}[m] + \frac{\mathbf{q}_R[m]-\tilde{\mathbf{w}}_{E_n}}{\|\mathbf{q}_R[m]-\tilde{\mathbf{w}}_{E_n}\|}\Delta Q_n$, respectively. Similarly, the power gain $\grave{g}_{JE_n}[m]$ of the worst channel from U_R-E_n is acquired at $\mathbf{w}_{JE_n}[m] = \tilde{\mathbf{w}}_{JE_n}[m] - \frac{\mathbf{q}_J[m]-\tilde{\mathbf{w}}_{E_n}}{\|\mathbf{q}_J[m]-\tilde{\mathbf{w}}_{E_n}\|}\Delta Q_n$.

Finally, $\grave{g}_{SE_n} \triangleq \max_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} g_{SE_n}$, $\grave{g}_{RE_n}[m] \triangleq \max_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} g_{RE_n}[m]$, and $\grave{g}_{JE_n}[m] \triangleq \max_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} g_{JE_n}[m]$ are given as

$$\begin{aligned}\grave{g}_{SE_n} &= \frac{\rho_0}{((\|\mathbf{w}_S-\tilde{\mathbf{w}}_{E_n}\|-\Delta Q_n)^2)}\xi,\\ \grave{g}_{RE_n}[m] &= \frac{\rho_0}{((\|\mathbf{q}_R[m]-\tilde{\mathbf{w}}_{E_n}\|-\Delta Q_n)^2+H_R^2)}\xi,\\ \grave{g}_{JE_n}[m] &= \frac{\rho_0}{((\|\mathbf{q}_J[m]-\tilde{\mathbf{w}}_{E_n}\|+\Delta Q_n)^2+H_J^2)}\xi.\end{aligned} \tag{26}$$

According to Jensen's inequality, if $f(x)$ is concave with respect to x, one can have $\mathbb{E}[f(x)] \le f(\mathbb{E}[x])$ and it can be observed that $\left(1+\frac{\tilde{g}_{SE_n}P_S[m]}{\tilde{g}_{JE_n}[m]P_J[m]+\sigma_{E_n}^2}+\frac{\tilde{g}_{RE_n}[m]P_R[m]}{\tilde{g}_{JE_n}[m]P_J[m]+\sigma_{E_n}^2}\right)$ is concave with respect to ξ [16]. So, in (26), we can substitute $\mathbb{E}_\xi[\xi] = 1$, and the upper bound of eavesdropping rate. Therefore, we can get $\tilde{g}_{SE_n} = \frac{\rho_0}{((\|\mathbf{w}_S-\tilde{\mathbf{w}}_{E_n}\|-\Delta Q_n)^2)}$, $\tilde{g}_{RE_n}[m] = \frac{\rho_0}{((\|\mathbf{q}_R[m]-\tilde{\mathbf{w}}_{E_n}\|-\Delta Q_n)^2+H_R^2)}$, and $\tilde{g}_{JE_n}[m] = \frac{\rho_0}{((\|\mathbf{q}_J[m]-\tilde{\mathbf{w}}_{E_n}\|+\Delta Q_n)^2+H_J^2)}$. From these values, $\max_{\Delta x_n,\Delta y_n\in\mathbb{E}_n} R_{\rm wir}[m] \le R_{\rm wir}^{\rm ub}[m]$ can be obtained and given as

$$R_{\rm wir}^{\rm ub}[m] = \frac{(1-\alpha[m])\delta_m}{2}\log_2\left(1+\sum_{n=1}^{N}\left(\frac{\tilde{g}_{SE_n}P_S[m]}{\tilde{g}_{JE_n}[m]P_J[m]+\sigma_{E_n}^2} + \frac{\tilde{g}_{RE_n}[m]P_R[m]}{\tilde{g}_{JE_n}[m]P_J[m]+\sigma_{E_n}^2}\right)\right). \tag{27}$$

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APPENDIX B

As per the first-order Taylor expansion for a convex function and given feasible initial points $P_S^0[m]$ and $P_R^0[m]$, the global under-estimator of $-\log_2(1+\sum_{n=1}^{N}(b_1[m]P_S[m]+b_2[m]P_R[m]))$ for any given $P_S^{\psi}[m]$ and $P_R^{\psi}[m]$ at the ψ_{th} iteration is expressed as

$$\begin{aligned}&-\log_2\Big(1+\sum_{n=1}^{N}(b_1[m]P_S[m]+b_2[m]P_R[m])\Big)\\&\geq -\log_2\Big(1+\sum_{n=1}^{N}(b_1[m]P_S^{\psi}[m]+b_2[m]P_R^{\psi}[m])\Big)\\&-\sum_{n=1}^{N}\left(\frac{b_1[m](P_S[m]-P_S^{\psi}[m])}{1+b_1[m]P_S^{\psi}[m]+b_2[m]P_R^{\psi}[m]}\right)\\&-\sum_{n=1}^{N}\left(\frac{b_2[m](P_R[m]-P_R^{\psi}[m])}{1+b_1[m]P_S^{\psi}[m]+b_2[m]P_R^{\psi}[m]}\right).\end{aligned}\tag{28}$$

where $b_1[m] = \tilde{g}_{SE_n}/(\tilde{g}_{JE_n}[m]P_J[m]+\sigma_{E_n}^2)$, $b_2[m] = \tilde{g}_{RE_n}[m]/(\tilde{g}_{JE_n}[m]P_J[m]+\sigma_{E_n}^2)$. Thus, after substituting (28) in (11), and taking $B_1^{\psi}[m] = b_1[m]/(1+b_1[m]P_S^{\psi}[m]+b_2[m]P_R^{\psi}[m])$, $B_2^{\psi}[m] = b_2[m]/(1+b_1[m]P_S^{\psi}[m]+b_2[m]P_R^{\psi}[m])$, $B_3^{\psi}[m] = \sum_{n=1}^{N}\left(\frac{a_3[m]P_S^{\psi}[m]+a_4[m]P_R^{\psi}[m]}{(1+b_1[m]P_S^{\psi}[m]+b_2[m]P_R^{\psi}[m])}\right)-\log_2(1+\sum_{n=1}^{N}(b_1[m]P_S^{\psi}[m]+b_2[m]P_R^{\psi}[m]))$, $\alpha_f[m]=\frac{(1-\alpha[m])\delta_m}{2}$, (P3) can be obtained as

$$\text{(P5)}: \max_{P_S[m],P_R[m]} \frac{1}{M}\sum_{m=1}^{M} S_{v2}[m] \tag{29a}$$

$$\text{s.t. } S_{v2}[m]\leq \alpha_f[m]\log_2(1+a_1[m]P_S[m]) - \alpha_f[m]\sum_{n=1}^{N}\Big(B_1^{\psi}[m]P_S[m]-B_2^{\psi}[m]P_R[m]\Big)+B_3^{\psi}[m], \tag{29b}$$

$$S_{v2}[m]\leq \alpha_f[m]\log_2(1+b_1[m]P_R[m]) - \alpha_f[m]\sum_{n=1}^{N}\Big(B_1^{\psi}[m]P_S[m]-B_2^{\psi}[m]P_R[m]\Big)+B_3^{\psi}[m], \tag{29c}$$

$$\sum_{m=1}^{M}(\log_2(1+a_1[m]P_S[m])-\log_2(1+b_1[m]P_R[m]))\leq 0, \tag{29d}$$

(10b), (10c), (10e), (10f), (10g), (10h),

In (29d), the term $\log_2(1+a_1[m]P_S[m])$ is concave with respect to the variable $P_R[m]$. However, $-\log_2(1+b_1[m]P_R[m])$ is convex terms in regard to $P_R[m]$ and need to be handled. Therefore, we again utilize the first-order Taylor expansion method to tackle this and reformulate the (P5) into approximated problem (P5.1), expressed in (13a).

APPENDIX C

After substituting the auxiliary variables and slack variables, the U_R's trajectory subproblem is reformulated as (P7) given in (30a). Nevertheless, (P7) remains non-convex due to the non-convex constraints. It is noted that the right side of the constraints in (30b) and (30c) is in the form of convex-minus-convex. Therefore, we can utilize the SCA method to approximate the first term of the objective function of the problem (P7) with its corresponding concave lower bound.

$$\text{(P7)}: \max_{\substack{\mathbf{q}_R[m],S_{v4}[m]\\ \mathbb{L}_{SR}[m],\mathbb{M}_{RE}[m],\mathbb{N}_{RD}[m]}} \frac{1}{M}\sum_{m=1}^{M}\alpha_f[m]S_{v4}[m] \tag{30a}$$

$$\text{s.t. } S_{v4}[m]\leq\left(\log_2\left(1+\frac{D_1[m]}{\mathbb{L}_{SR}[m]}\right)-\log_2\left(1+\sum_{n=1}^{N}\frac{D_2[m]}{\mathbb{M}_{RE}[m]}\right)\right), \tag{30b}$$

$$S_{v4}[m]\leq\left(\log_2\left(1+\frac{D_3[m]}{\mathbb{N}_{RD}[m]}\right)-\log_2\left(1+\sum_{n=1}^{N}\frac{D_2[m]}{\mathbb{M}_{RE}[m]}\right)\right), \tag{30c}$$

$$\mathbb{L}_{SR}[m]\leq\|\mathbf{w}_S-\mathbf{q}_R[m]\|^2+H_R^2, \tag{30d}$$

$$\mathbb{M}_{RE}[m]\geq(\|\mathbf{q}_R[m]-\mathbf{w}_{E_n}\|+\Delta Q_n)^2+H_R^2, \tag{30e}$$

$$\mathbb{N}_{RD}[m]\leq\|\mathbf{q}_R[m]-\mathbf{w}_D\|^2+H_R^2, \tag{30f}$$

$$\sum_{m=1}^{M}\alpha_f[m]\left(\log_2\left(1+\frac{D_1[m]}{\mathbb{L}_{SR}[m]}\right)\right)-\left(\log_2\left(1+\frac{D_3[m]}{\mathbb{N}_{RD}[m]}\right)\right)\leq 0, \tag{30g}$$

(10j), $\cdots$, (10m).

For the given local points, $L_{SR}^{\psi}[m]$ and $N_{RD}^{\psi}[m]$, first-order Taylor approximation of the convex term are given by $\log_2(1+\frac{D_1[m]}{\mathbb{L}_{SR}[m]})\geq D_4^{\psi}[m]\mathbb{L}_{SR}[m]+D_5^{\psi}[m]$ and $\log_2(1+\frac{D_6[m]}{\mathbb{N}_{RD}[m]})\geq D_6^{\psi}[m]\mathbb{N}_{RD}[m]+D_7^{\psi}[m]$, where $D_4^{\psi}[m]=\frac{-D_1[m]\ln 2}{L_{SR}^{\psi}[m](D_1[m]+L_{SR}^{\psi}[m])}$, $D_5^{\psi}[m]=\log_2(1+\frac{D_4[m]}{L_{SR}^{\psi}[m]})+\frac{-D_1[m]\ln 2L_{SR}^{\psi}[m]}{L_{SR}^{\psi}[m](D_1[m]+L_{SR}^{\psi}[m])}$, $D_6^{\psi}[m]=\frac{-D_3[m]\ln 2}{N_{RD}^{\psi}[m](D_3[m]+N_{RD}^{\psi}[m])}$, and $D_7^{\psi}[m]=\log_2(1+\frac{D_3[m]}{N_{RD}^{\psi}[m]})+\frac{-D_3[m]\ln 2N_{rd}^{\psi}[m]}{N_{RD}^{\psi}[m](D_3[m]+N_{RD}^{\psi}[m])}$. Furthermore, we again utilize SCA technique to obtain the lower bound or upper bound of the norm-squared function in (30d), (30e), (30f), and (10l) as

$$-\|\mathbf{q}_R[m]-\mathbf{w}_S\|^2\leq\|\mathbf{q}_R^{\psi}[m]-\mathbf{w}_S\|^2+2(\mathbf{q}_R^{\psi}[m]-\mathbf{w}_S)^T(\mathbf{q}_R[m]-\mathbf{q}_R^{\psi}[m])\triangleq f_1(\mathbf{q}_R[m],\mathbf{q}_R^{\psi}[m]). \tag{31}$$

Similarly, we can get $\|\mathbf{q}_R[m]-\mathbf{w}_D\|^2\geq\|\mathbf{q}_R^{\psi}[m]-\mathbf{w}_D\|^2+2(\mathbf{q}_R^{\psi}[m]-\mathbf{w}_D)^T(\mathbf{q}_R[m]-\mathbf{q}_R^{\psi}[m])\triangleq f_2(\mathbf{q}_R[m],\mathbf{q}_R^{\psi}[m])$, and $\|\mathbf{q}_R[m]-\mathbf{q}_J[m]\|^2\geq\|\mathbf{q}_R^{\psi}[m]-\mathbf{q}_J[m]\|^2+2(\mathbf{q}_R^{\psi}[m]-\mathbf{q}_J[m])^T(\mathbf{q}_R[m]-\mathbf{q}_R^{\psi}[m])\triangleq f_3(\mathbf{q}_R[m],\mathbf{q}_R^{\psi}[m])$. Now, we substitute these approximated values in (P7) and express the reformulated subproblem (P7.1) in (21a).

APPENDIX D

Similar to previous sections, we introduce a slack variable, $S_{v5}[m]$, to tackle the objective function. Moreover, we introduce the slack variable $\mathbb{O}_{JE}[m]=\{O_{JE}[1],\ldots,O_{JE}[m]\}$ to relax the non-convex constraints. We substitute $G_1[m]=g_{SR}[m]P_S[m]$, $G_2[m]=g_{RD}[m]P_R[m]$, and $G_3=\tilde{g}_{SE_n}P_S[m]+\tilde{g}_{RE_n}[m]P_R[m]$. Accordingly, we formulate the subproblem as (32a). However, the constraints in (32b), (32c), and (32d) are still non-convex, i.e., (P8) is a non-convex problem. Similar to the previous subsections, ae adopt SCA method and employ Taylor approximation of the convex term in (32b) and (32c).

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$$\text{(P8)}: \max_{\mathbf{q}_H[m], S_{v5}[m]\mathbb{O}_{JE}[m]} \frac{1}{M}\sum_{m=1}^{M}\alpha_f[m]S_{v5}[m] \tag{32a}$$

$$\text{s.t.}\quad S_{v5}[m] \le \log_2(1+G_1[m]) - \log_2\left(1+\sum_{n=1}^{N}\frac{G_3}{\frac{\rho_0 P_J[m]}{\mathbb{O}_{JE}[m]}+\sigma_{E_n}^2}\right), \tag{32b}$$

$$S_{v5}[m] \le \log_2(1+G_2[m]) - \log_2\left(1+\sum_{n=1}^{N}\frac{G_3}{\frac{\rho_0 P_J[m]}{\mathbb{O}_{JE}[m]}+\sigma_{E_n}^2}\right), \tag{32c}$$

$$\mathbb{O}_{JE}[m] \ge (\|\mathbf{q}_J[m]-\mathbf{w}_{E_n}\|+\Delta Q_n)^2 + H_J^2, \tag{32d}$$

$$\|\mathbf{q}_R[m]-\mathbf{q}_J[m]\|^2 \ge d_{\min} \quad \forall m, \tag{32e}$$

$$\text{(10j)},\ \cdots,\ \text{(10k)},\ \text{(10m)}.$$

Consequently, $-\log_2\left(1+\sum_{n=1}^{N}\frac{G_3}{\frac{\rho_0 P_J[m]}{\mathbb{O}_{JE}[m]}+\sigma_{E_n}^2}\right) \ge G_6^{\psi}[m] - \mathbb{O}_{JE}[m]$, where $G_4^{\psi}[m] = \frac{\rho_0 O_{JE}^{\psi}[m] G_3[m] P_J[m]}{\ln 2(\rho_0 P_J[m]+\sigma_{E_n}^2 O_{JE}^{\psi}[m])(\rho_0 P_J[m]+(\sigma_{E_n}^2+G_3)O_{JE}^{\psi}[m])}$, $G_5^{\psi}[m] = \log(1+\sum_{n=1}^{N}\frac{G_3 O_{JE}^{\psi}[m]}{\rho_0 P_J[m]+\sigma_{E_n}^2 O_{JE}^{\psi}[m]})$, and $G_6^{\psi}[m] = -\sum_{n=1}^{N}(G_4^{\psi}[m]) - G_5^{\psi}[m]$. The non-convex conditions of (32b) and (32c) are relaxed. Now, we tackle the left-hand side terms of (32e) by applying the first order Taylor expansion to derive their lower bound as $\|\mathbf{q}_J[m]-\mathbf{q}_R[m]\|^2 \ge \mathbf{q}_J^{\psi}[m]-\mathbf{q}_R[m] + 2(\mathbf{q}_J^{\psi}[m]-\mathbf{q}_R[m])^T(\mathbf{q}_J[m]-\mathbf{q}_J^{\psi}[m] \triangleq f_4(\mathbf{q}_J[m],\mathbf{q}_J^{\psi}[m]))$. Finally, we substitute these approximated values in (P8) and express (P8.1) in (22a).

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